lab-automation-twin (AICOR OKF module): description-logic axioms

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Names are AICOR L2 names; dul:/soma: are AICOR's upper-level stand-ins; superclasses not defined here are upstream (AICOR core, robot-scientist, VRB, normative-admissibility).

Properties, upper-level placements and the SDT node / referent split

dul:PhysicalPlace (stand-in) a place with a physical extent; its sub-places are places (DUL: PhysicalPlace ⊑ PhysicalObject).

$$\mathsf{dul{:}PhysicalPlace} \sqsubseteq \mathsf{dul{:}PhysicalObject}$$
$$\mathsf{dul{:}PhysicalPlace} \sqsubseteq \exists \mathit{hasSubPlace}.\mathsf{dul{:}PhysicalPlace}$$

LabArtifact A designed physical object of the lab that the twin represents.

$$\mathsf{LabArtifact} \sqsubseteq \mathsf{dul{:}PhysicalObject}$$
$$\mathsf{LabArtifact} \sqsubseteq \exists \mathit{isRepresentedBy}.\mathsf{LabSDTNode}$$

LabSDTNode A node of the lab's semantic digital twin (one USD prim): information that represents one entity.

$$\mathsf{LabSDTNode} \sqsubseteq \mathsf{SceneNode}$$
$$\mathsf{LabSDTNode} \sqsubseteq \exists \mathit{representsEntity}.\mathsf{Entity}$$
$$\mathsf{LabSDTNode} \sqsubseteq \exists \mathit{usdPrimPath}.\mathsf{xsd{:}string}$$

GroupingNode A structural SDT node (USD scope / xform) that groups child nodes; it represents the grouping only.

$$\mathsf{GroupingNode} \sqsubseteq \mathsf{LabSDTNode}$$
$$\mathsf{GroupingNode} \sqsubseteq \exists \mathit{hasChildNode}.\mathsf{LabSDTNode}$$

LabSemanticDigitalTwin The semantic digital twin of a laboratory site: the belief state about it, structured as a USD scene graph.

$$\mathsf{LabSemanticDigitalTwin} \sqsubseteq \mathsf{SemanticDigitalTwin}$$
$$\mathsf{LabSemanticDigitalTwin} \sqsubseteq \exists \mathit{hasNodeStructure}.\mathsf{LabSceneHierarchy}$$

LabSceneHierarchy The USD namespace hierarchy of the lab twin (lab_twin.usda + culture layer).

$$\mathsf{LabSceneHierarchy} \sqsubseteq \mathsf{HierarchicalSceneStructure}$$
$$\mathsf{LabSceneHierarchy} \sqsubseteq \exists \mathit{hasNode}.\mathsf{LabSDTNode}$$

SimulationSettings Physics or rendering settings of the twin (physics scene, sky and sun lights).

$$\mathsf{SimulationSettings} \sqsubseteq \mathsf{dul{:}Description}$$
$$\mathsf{SimulationSettings} \sqsubseteq \exists \mathit{configures}.\mathsf{LabSemanticDigitalTwin}$$

SimulationSettingsNode An SDT node holding settings.

$$\mathsf{SimulationSettingsNode} \sqsubseteq \mathsf{LabSDTNode}$$
$$\mathsf{SimulationSettingsNode} \sqsubseteq \exists \mathit{representsEntity}.\mathsf{SimulationSettings}$$

RenderRepresentationNode A render-only USD prim (looks, materials, collision / visual geometry): representation, not SDT knowledge.

$$\mathsf{RenderRepresentationNode} \sqsubseteq \mathsf{SceneNode}$$
$$\mathsf{RenderRepresentationNode} \sqsubseteq \neg \mathsf{LabSDTNode}$$

Laboratory site, premises, rooms, zones, building elements and fit-out

AutomatedBiologyLab A laboratory site that comprises at least one robot experiment cell (AICOR Lab 1).

$$\mathsf{AutomatedBiologyLab} \sqsubseteq \mathsf{LaboratorySite}$$
$$\mathsf{AutomatedBiologyLab} \sqsubseteq \exists \mathit{comprisesCell}.\mathsf{RobotExperimentCell}$$
$$\mathsf{AutomatedBiologyLab} \sqsubseteq \exists \mathit{locatedAt}.\mathsf{LabPremises}$$

LabPremises The floor area of a laboratory site.

$$\mathsf{LabPremises} \sqsubseteq \mathsf{dul{:}PhysicalPlace}$$
$$\mathsf{LabPremises} \sqsubseteq \exists \mathit{hasSubPlace}.\mathsf{LabRoom}$$

LabRoom A room of the premises.

$$\mathsf{LabRoom} \sqsubseteq \mathsf{dul{:}PhysicalPlace}$$
$$\mathsf{LabRoom} \sqsubseteq \exists \mathit{isSubPlaceOf}.\mathsf{LabPremises}$$

LabHall The hall in which automation cells stand.

$$\mathsf{LabHall} \sqsubseteq \mathsf{LabRoom}$$
$$\mathsf{LabHall} \sqsubseteq \exists \mathit{hostsObject}.\mathsf{LaboratoryCell}$$

StorageRoom A room for consumables, reagents and cold storage (ST01).

$$\mathsf{StorageRoom} \sqsubseteq \mathsf{LabRoom}$$
$$\mathsf{StorageRoom} \sqsubseteq \exists \mathit{hostsObject}.(\mathsf{ColdStorageDevice} \sqcup \mathsf{StorageShelving})$$

LabZone A functional area (layout S1).

$$\mathsf{LabZone} \sqsubseteq \mathsf{dul{:}PhysicalPlace}$$
$$\mathsf{LabZone} \sqsubseteq \exists \mathit{isSubPlaceOf}.(\mathsf{LabRoom} \sqcup \mathsf{LabPremises})$$

EntranceZone Entrance and gowning.

$$\mathsf{EntranceZone} \sqsubseteq \mathsf{LabZone}$$
$$\mathsf{EntranceZone} \sqsubseteq \exists \mathit{hostsObject}.\mathsf{PPEStation}$$
$$\mathsf{EntranceZone} \sqcap \mathsf{OperatorArea} \sqsubseteq \bot$$

OperatorArea Monitoring and engineering desks.

$$\mathsf{OperatorArea} \sqsubseteq \mathsf{LabZone}$$
$$\mathsf{OperatorArea} \sqsubseteq \exists \mathit{hostsObject}.\mathsf{OperatorWorkstation}$$
$$\mathsf{OperatorArea} \sqcap \mathsf{EntranceZone} \sqsubseteq \bot$$

CulturePrepArea Manual culture work upstream.

$$\mathsf{CulturePrepArea} \sqsubseteq \mathsf{LabZone}$$
$$\mathsf{CulturePrepArea} \sqsubseteq \exists \mathit{hostsObject}.\mathsf{BiosafetyCabinet}$$
$$\mathsf{CulturePrepArea} \sqcap \mathsf{EntranceZone} \sqsubseteq \bot$$

WashWasteArea Sterilisation and waste.

$$\mathsf{WashWasteArea} \sqsubseteq \mathsf{LabZone}$$
$$\mathsf{WashWasteArea} \sqsubseteq \exists \mathit{hostsObject}.(\mathsf{Autoclave} \sqcup \mathsf{WasteStation})$$
$$\mathsf{WashWasteArea} \sqcap \mathsf{EntranceZone} \sqsubseteq \bot$$

CellZone Floor envelope of an automation cell.

$$\mathsf{CellZone} \sqsubseteq \mathsf{LabZone}$$
$$\mathsf{CellZone} \sqsubseteq \exists \mathit{hostsObject}.\mathsf{LaboratoryCell}$$
$$\mathsf{CellZone} \sqcap \mathsf{EntranceZone} \sqsubseteq \bot$$

ServiceCorridor Technician access to the rear instruments.

$$\mathsf{ServiceCorridor} \sqsubseteq \mathsf{LabZone}$$
$$\mathsf{ServiceCorridor} \sqsubseteq \exists \mathit{adjoinsZone}.\mathsf{CellZone}$$
$$\mathsf{ServiceCorridor} \sqcap \mathsf{EntranceZone} \sqsubseteq \bot$$

LogisticsArea The AMR lane and walkway to the cell portal.

$$\mathsf{LogisticsArea} \sqsubseteq \mathsf{LabZone}$$
$$\mathsf{LogisticsArea} \sqsubseteq \exists \mathit{adjoinsZone}.\mathsf{CellZone}$$
$$\mathsf{LogisticsArea} \sqcap \mathsf{EntranceZone} \sqsubseteq \bot$$

LabCeiling Ceiling with services.

$$\mathsf{LabCeiling} \sqsubseteq \mathsf{dul{:}PhysicalPlace}$$
$$\mathsf{LabCeiling} \sqsubseteq \exists \mathit{hostsObject}.\mathsf{BuildingServiceDevice}$$

LabBuildingShell Walls, doors, windows and services.

$$\mathsf{LabBuildingShell} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{LabBuildingShell} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{LabDoor}$$

LabDoor A door with a moving leaf.

$$\mathsf{LabDoor} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{LabDoor} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{DoorLeaf}$$

HingedDoor Leaf swings on a hinge.

$$\mathsf{HingedDoor} \sqsubseteq \mathsf{LabDoor}$$
$$\mathsf{HingedDoor} \sqsubseteq \exists \mathit{hasDirectComponent}.\exists \mathit{hasMechanismJointType}.\mathsf{Revolute}$$
$$\mathsf{HingedDoor} \sqcap \mathsf{SlidingDoor} \sqsubseteq \bot$$

SlidingDoor Leaf slides.

$$\mathsf{SlidingDoor} \sqsubseteq \mathsf{LabDoor}$$
$$\mathsf{SlidingDoor} \sqsubseteq \exists \mathit{hasDirectComponent}.\exists \mathit{hasMechanismJointType}.\mathsf{Prismatic}$$
$$\mathsf{SlidingDoor} \sqcap \mathsf{HingedDoor} \sqsubseteq \bot$$

InterlockedDoor A door whose open state gates safe robot motion.

$$\mathsf{InterlockedDoor} \sqsubseteq \mathsf{LabDoor}$$
$$\mathsf{InterlockedDoor} \sqsubseteq \exists \mathit{isMonitoredBy}.\mathsf{DoorInterlockSwitch}$$

DoorLeaf The moving panel of a door.

$$\mathsf{DoorLeaf} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{DoorLeaf} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{LabDoor}$$

DoorFrame The fixed frame of a door.

$$\mathsf{DoorFrame} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{DoorFrame} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{LabDoor}$$

AMRRoute A navigation route of an AMR between docks (a USD curve).

$$\mathsf{AMRRoute} \sqsubseteq \mathsf{SpatialPose}$$
$$\mathsf{AMRRoute} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{NavigationTask}$$
$$\mathsf{AMRRoute} \sqsubseteq \exists \mathit{startsAt}.\mathsf{Dock}$$
$$\mathsf{AMRRoute} \sqsubseteq \exists \mathit{endsAt}.\mathsf{Dock}$$

RobotExperimentCell An automation cell of robots and automated stations behind a safety enclosure (Cell 01).

$$\mathsf{RobotExperimentCell} \sqsubseteq \mathsf{LaboratoryCell}$$
$$\mathsf{RobotExperimentCell} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{CellRobot}$$
$$\mathsf{RobotExperimentCell} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{LabAutomationDevice}$$
$$\mathsf{RobotExperimentCell} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{SafetyEnclosure}$$

AutomationCell Defined: a laboratory cell with robots and automated devices as direct components.

$$\mathsf{AutomationCell} \sqsubseteq \mathsf{LaboratoryCell}$$
$$\mathsf{AutomationCell} \equiv \mathsf{LaboratoryCell} \sqcap \exists \mathit{hasDirectComponent}.\mathsf{Robot} \sqcap \exists \mathit{hasDirectComponent}.\mathsf{LabAutomationDevice}$$

SafetyEnclosure Guard enclosure bounding the robot envelope: glazed sides, interlocked doors, e-stops (spec §4.1).

$$\mathsf{SafetyEnclosure} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{SafetyEnclosure} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{LabDoor}$$
$$\mathsf{SafetyEnclosure} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{EmergencyStop}$$

AMRPortal Opening in the enclosure for the sample AMR, guarded by a muted light curtain.

$$\mathsf{AMRPortal} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{AMRPortal} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{SafetyEnclosure}$$
$$\mathsf{AMRPortal} \sqsubseteq \exists \mathit{isMonitoredBy}.\mathsf{LightCurtain}$$

HMIPanel Operator touch panel.

$$\mathsf{HMIPanel} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{HMIPanel} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{LabArtifact}$$

Worktop The working surface of a bench.

$$\mathsf{Worktop} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{Worktop} \sqsubseteq \exists \mathit{isDirectComponentOf}.(\mathsf{AutomationBench} \sqcup \mathsf{LabBench})$$

AutomationBench Machine bench carrying stations at transfer height (spec §4.2).

$$\mathsf{AutomationBench} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{AutomationBench} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{Worktop}$$
$$\mathsf{AutomationBench} \sqsubseteq \exists \mathit{supportsObject}.\mathsf{LabAutomationDevice}$$

StationSupport Plinth or side table carrying a station at its transfer height.

$$\mathsf{StationSupport} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{StationSupport} \sqsubseteq \exists \mathit{supportsObject}.\mathsf{LabArtifact}$$

VibrationIsolatedSupport Support decoupled from the bench (imaging sub-base).

$$\mathsf{VibrationIsolatedSupport} \sqsubseteq \mathsf{StationSupport}$$
$$\mathsf{VibrationIsolatedSupport} \sqsubseteq \exists \mathit{supportsObject}.\mathsf{ImagingDevice}$$

RailCover Cover over a robot rail.

$$\mathsf{RailCover} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{RailCover} \sqsubseteq \exists \mathit{coversObject}.\mathsf{LinearRail}$$

StationHood Glazed hood over a station deck.

$$\mathsf{StationHood} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{StationHood} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{LabAutomationDevice}$$

FitOutAssembly The cell's furniture and covers as installed.

$$\mathsf{FitOutAssembly} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{FitOutAssembly} \sqsubseteq \exists \mathit{hasDirectComponent}.(\mathsf{StationSupport} \sqcup \mathsf{AutomationBench} \sqcup \mathsf{RailCover})$$

SafetyDevice Equipment whose function is safety.

$$\mathsf{SafetyDevice} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{SafetyDevice} \sqsubseteq \exists \mathit{mitigatesHazard}.\mathsf{LabHazard}$$

EmergencyStop Latching e-stop button.

$$\mathsf{EmergencyStop} \sqsubseteq \mathsf{SafetyDevice}$$
$$\mathsf{EmergencyStop} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{EmergencyStop} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{LabArtifact}$$

LightCurtain Optical safety curtain across the portal; muting admits the AMR only.

$$\mathsf{LightCurtain} \sqsubseteq \mathsf{SafetyDevice}$$
$$\mathsf{LightCurtain} \sqsubseteq \mathsf{DeviceSensor}$$
$$\mathsf{LightCurtain} \sqsubseteq \exists \mathit{monitorsEntity}.\mathsf{AMRPortal}$$

DoorInterlockSwitch Door switch that withholds safe motion while the door is open.

$$\mathsf{DoorInterlockSwitch} \sqsubseteq \mathsf{SafetyDevice}$$
$$\mathsf{DoorInterlockSwitch} \sqsubseteq \mathsf{DeviceSensor}$$
$$\mathsf{DoorInterlockSwitch} \sqsubseteq \exists \mathit{monitorsEntity}.\mathsf{LabDoor}$$

FireSafetyDevice Fire detection or suppression.

$$\mathsf{FireSafetyDevice} \sqsubseteq \mathsf{SafetyDevice}$$
$$\mathsf{FireSafetyDevice} \sqsubseteq \exists \mathit{mitigatesHazard}.\mathsf{FireHazard}$$

SprinklerHead Pendent sprinkler head.

$$\begin{aligned}\mathsf{SprinklerHead} &\sqsubseteq \mathsf{FireSafetyDevice}\\ \mathsf{SprinklerHead} &\sqsubseteq \exists \mathit{locatedAt}.\mathsf{LabCeiling}\\ \mathsf{SprinklerHead} \sqcap \mathsf{SmokeDetector} &\sqsubseteq \bot\end{aligned}$$

SmokeDetector Ceiling smoke detector.

$$\begin{aligned}\mathsf{SmokeDetector} &\sqsubseteq \mathsf{FireSafetyDevice}\\ \mathsf{SmokeDetector} &\sqsubseteq \mathsf{DeviceSensor}\\ \mathsf{SmokeDetector} &\sqsubseteq \exists \mathit{locatedAt}.\mathsf{LabCeiling}\\ \mathsf{SmokeDetector} \sqcap \mathsf{SprinklerHead} &\sqsubseteq \bot\end{aligned}$$

FireExtinguisher Portable extinguisher.

$$\begin{aligned}\mathsf{FireExtinguisher} &\sqsubseteq \mathsf{FireSafetyDevice}\\ \mathsf{FireExtinguisher} &\sqsubseteq \exists \mathit{locatedAt}.\mathsf{dul{:}PhysicalPlace}\\ \mathsf{FireExtinguisher} \sqcap \mathsf{SprinklerHead} &\sqsubseteq \bot\end{aligned}$$

FirstAidBox First-aid cabinet.

$$\begin{aligned}\mathsf{FirstAidBox} &\sqsubseteq \mathsf{SafetyDevice}\\ \mathsf{FirstAidBox} &\sqsubseteq \exists \mathit{locatedAt}.\mathsf{dul{:}PhysicalPlace}\end{aligned}$$

EmergencyShower Drench shower.

$$\begin{aligned}\mathsf{EmergencyShower} &\sqsubseteq \mathsf{SafetyDevice}\\ \mathsf{EmergencyShower} &\sqsubseteq \exists \mathit{mitigatesHazard}.\mathsf{ChemicalExposureHazard}\end{aligned}$$

O2DepletionMonitor Oxygen-depletion monitor guarding cryogen / gas storage.

$$\begin{aligned}\mathsf{O2DepletionMonitor} &\sqsubseteq \mathsf{SafetyDevice}\\ \mathsf{O2DepletionMonitor} &\sqsubseteq \mathsf{DeviceSensor}\\ \mathsf{O2DepletionMonitor} &\sqsubseteq \exists \mathit{mitigatesHazard}.\mathsf{AsphyxiationHazard}\end{aligned}$$

AccessControlReader Card reader releasing a door.

$$\begin{aligned}\mathsf{AccessControlReader} &\sqsubseteq \mathsf{SafetyDevice}\\ \mathsf{AccessControlReader} &\sqsubseteq \exists \mathit{monitorsEntity}.\mathsf{LabDoor}\end{aligned}$$

BuildingServiceDevice Technical building equipment.

$$\begin{aligned}\mathsf{BuildingServiceDevice} &\sqsubseteq \mathsf{LabArtifact}\\ \mathsf{BuildingServiceDevice} &\sqsubseteq \exists \mathit{locatedAt}.\mathsf{dul{:}PhysicalPlace}\end{aligned}$$

HVACDiffuser Supply-air diffuser.

$$\begin{aligned}\mathsf{HVACDiffuser} &\sqsubseteq \mathsf{BuildingServiceDevice}\\ \mathsf{HVACDiffuser} &\sqsubseteq \exists \mathit{locatedAt}.\mathsf{LabCeiling}\end{aligned}$$

CeilingLuminaire LED ceiling panel.

$$\begin{aligned}\mathsf{CeilingLuminaire} &\sqsubseteq \mathsf{BuildingServiceDevice}\\ \mathsf{CeilingLuminaire} &\sqsubseteq \exists \mathit{locatedAt}.\mathsf{LabCeiling}\end{aligned}$$

ServerRack Rack hosting the twin's compute.

$$\begin{aligned}\mathsf{ServerRack} &\sqsubseteq \mathsf{BuildingServiceDevice}\\ \mathsf{ServerRack} &\sqsubseteq \exists \mathit{hostsSoftware}.\mathsf{LabSemanticDigitalTwin}\end{aligned}$$

GloveDispenser Glove box holder.

$$\begin{aligned}\mathsf{GloveDispenser} &\sqsubseteq \mathsf{LabArtifact}\\ \mathsf{GloveDispenser} &\sqsubseteq \exists \mathit{locatedAt}.\mathsf{LabZone}\end{aligned}$$

LabCoatRack Coat hooks.

$$\begin{aligned}\mathsf{LabCoatRack} &\sqsubseteq \mathsf{LabArtifact}\\ \mathsf{LabCoatRack} &\sqsubseteq \exists \mathit{locatedAt}.\mathsf{EntranceZone}\end{aligned}$$

PPEStation Defined: a PPE station.

$$\mathsf{PPEStation} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{PPEStation} \equiv \mathsf{GloveDispenser} \sqcup \mathsf{LabCoatRack}$$

PottedPlant Decor (a plant is not a designed artifact).

$$\mathsf{PottedPlant} \sqsubseteq \mathsf{dul{:}PhysicalObject}$$
$$\mathsf{PottedPlant} \sqsubseteq \exists \mathit{hasPhysicalPart}.\mathsf{BiologicalSpecimen}$$
$$\mathsf{PottedPlant} \sqsubseteq \neg\mathsf{LabArtifact}$$

Whiteboard Wall whiteboard.

$$\mathsf{Whiteboard} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{Whiteboard} \sqsubseteq \exists \mathit{locatedAt}.\mathsf{LabZone}$$

LabBench Laboratory bench.

$$\mathsf{LabBench} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{LabBench} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{Worktop}$$

AMRDispatchBench Bench where staff stage AMR carriers (S1).

$$\mathsf{AMRDispatchBench} \sqsubseteq \mathsf{LabBench}$$
$$\mathsf{AMRDispatchBench} \sqsubseteq \exists \mathit{locatedAt}.\mathsf{StorageRoom}$$

StorageShelving Shelving with addressable locations.

$$\mathsf{StorageShelving} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{StorageShelving} \sqsubseteq \exists \mathit{locatedAt}.\mathsf{dul{:}PhysicalPlace}$$

PipetteCarousel Manual pipette stand.

$$\mathsf{PipetteCarousel} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{PipetteCarousel} \sqsubseteq \exists \mathit{locatedAt}.\mathsf{LabZone}$$

LabSinkUnit Sink with tap and eyewash.

$$\mathsf{LabSinkUnit} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{LabSinkUnit} \sqsubseteq \exists \mathit{hasEquipmentFunction}.\mathsf{WashingFunction}$$

WasteStation Segregated waste and sharps bins.

$$\mathsf{WasteStation} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{WasteStation} \sqsubseteq \exists \mathit{locatedAt}.\mathsf{WashWasteArea}$$

OperatorWorkstation Desk with monitors.

$$\mathsf{OperatorWorkstation} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{OperatorWorkstation} \sqsubseteq \exists \mathit{locatedAt}.\mathsf{OperatorArea}$$

LabStool Lab stool.

$$\mathsf{LabStool} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{LabStool} \sqsubseteq \exists \mathit{locatedAt}.\mathsf{LabZone}$$

Laboratory devices, automated stations, mechanisms and device sensors

PlateStorageFunction Store / buffer plates.

$$\mathsf{PlateStorageFunction} \sqsubseteq \mathsf{DeviceFunction}$$
$$\mathsf{PlateStorageFunction} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{PlateStoringTask}$$

QuarantineFunction Contain suspect material.

$$\mathsf{QuarantineFunction} \sqsubseteq \mathsf{DeviceFunction}$$
$$\mathsf{QuarantineFunction} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{QuarantiningTask}$$

CalibrationFunction Validate a measurement chain.

$$\mathsf{CalibrationFunction} \sqsubseteq \mathsf{DeviceFunction}$$
$$\mathsf{CalibrationFunction} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{CalibratingTask}$$

IdentificationFunction Read labware identity.

$$\mathsf{IdentificationFunction} \sqsubseteq \mathsf{DeviceFunction}$$
$$\mathsf{IdentificationFunction} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{IdentifyingTask}$$

ColdStorageFunction Keep material cold.

$$\mathsf{ColdStorageFunction} \sqsubseteq \mathsf{DeviceFunction}$$
$$\mathsf{ColdStorageFunction} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{ColdStoringTask}$$

SterilizationFunction Sterilise material.

$$\mathsf{SterilizationFunction} \sqsubseteq \mathsf{DeviceFunction}$$
$$\mathsf{SterilizationFunction} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{SterilizingTask}$$

AsepticWorkFunction Protect open cultures.

$$\mathsf{AsepticWorkFunction} \sqsubseteq \mathsf{DeviceFunction}$$
$$\mathsf{AsepticWorkFunction} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{AsepticHandlingTask}$$

TemperatureControlFunction Hold a sample at temperature.

$$\mathsf{TemperatureControlFunction} \sqsubseteq \mathsf{DeviceFunction}$$
$$\mathsf{TemperatureControlFunction} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{TemperatureControllingTask}$$

PlateStorageDevice Defined: an automated device that stores plates in a cassette.

$$\mathsf{PlateStorageDevice} \sqsubseteq \mathsf{LabAutomationDevice}$$
$$\mathsf{PlateStorageDevice} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{PlateStorageDevice} \equiv \mathsf{LabAutomationDevice} \sqcap \exists \mathit{hasDirectComponent}.\mathsf{PlateCassette}$$
$$\mathsf{PlateStorageDevice} \sqcap \mathsf{DispensingDevice} \sqsubseteq \bot$$

AutomatedPlateHotel Ambient automated plate store (L09, LiCONiC LPE220 baseline).

$$\mathsf{AutomatedPlateHotel} \sqsubseteq \mathsf{PlateStorageDevice}$$
$$\mathsf{AutomatedPlateHotel} \sqsubseteq \exists \mathit{hasDeviceFunction}.\mathsf{PlateStorageFunction}$$
$$\mathsf{AutomatedPlateHotel} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{PlateShuttle}$$
$$\mathsf{AutomatedPlateHotel} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{InternalPlateHandler}$$
$$\mathsf{AutomatedPlateHotel} \sqsubseteq \neg\exists \mathit{hasDirectComponent}.\mathsf{IncubationChamber}$$
$$\mathsf{AutomatedPlateHotel} \sqcap \mathsf{AutomatedCO2Incubator} \sqsubseteq \bot$$

AutomatedCO2Incubator Robot-accessible CO2 incubator (L10, LiCONiC STX44-SA baseline).

$$\mathsf{AutomatedCO2Incubator} \sqsubseteq \mathsf{IncubationDevice}$$
$$\mathsf{AutomatedCO2Incubator} \sqsubseteq \mathsf{PlateStorageDevice}$$
$$\mathsf{AutomatedCO2Incubator} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{IncubationChamber}$$
$$\mathsf{AutomatedCO2Incubator} \sqsubseteq \exists \mathit{declaresSetpoint}.\mathsf{OperatingSetpoint}$$
$$\mathsf{AutomatedCO2Incubator} \sqcap \mathsf{AutomatedPlateHotel} \sqsubseteq \bot$$

AutomatedLiquidHandler Gantry liquid handler (L12, Hamilton ZEUS LT baseline).

$$\mathsf{AutomatedLiquidHandler} \sqsubseteq \mathsf{DispensingDevice}$$
$$\mathsf{AutomatedLiquidHandler} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{AutomatedLiquidHandler} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{GantryAxis}$$
$$\mathsf{AutomatedLiquidHandler} \sqsubseteq \exists \mathit{hasPhysicalPart}.\mathsf{PipettingChannel}$$

BrightFieldImager Bright-field imager (L17, OpenFlexure-derived).

$$\mathsf{BrightFieldImager} \sqsubseteq \mathsf{ImagingDevice}$$
$$\mathsf{BrightFieldImager} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{BrightFieldImager} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{MotorizedStage}$$
$$\mathsf{BrightFieldImager} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{FocusDrive}$$
$$\mathsf{BrightFieldImager} \sqsubseteq \exists \mathit{hasPhysicalPart}.\mathsf{ObjectiveTurret}$$

QCCalibrationStation QC / calibration bench with reference artifacts (L21).

$$\mathsf{QCCalibrationStation} \sqsubseteq \mathsf{LabAutomationDevice}$$
$$\mathsf{QCCalibrationStation} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{QCCalibrationStation} \sqsubseteq \exists \mathit{hasDeviceFunction}.\mathsf{CalibrationFunction}$$
$$\mathsf{QCCalibrationStation} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{LightBarrierSensor}$$
$$\mathsf{QCCalibrationStation} \sqcap \mathsf{PlateStorageDevice} \sqsubseteq \bot$$

QuarantineStation Lockable quarantine store (H4).

$$\begin{aligned}
&\mathsf{QuarantineStation} \sqsubseteq \mathsf{LabAutomationDevice}\\
&\mathsf{QuarantineStation} \sqsubseteq \mathsf{LabArtifact}\\
&\mathsf{QuarantineStation} \sqsubseteq \exists \mathit{hasDeviceFunction}.\mathsf{QuarantineFunction}\\
&\mathsf{QuarantineStation} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{LockableDoor}\\
&\mathsf{QuarantineStation} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{KeySwitch}\\
&\mathsf{QuarantineStation} \sqcap \mathsf{PlateStorageDevice} \sqsubseteq \bot
\end{aligned}$$

Dock A place an AMR docks at.

$$\begin{aligned}
&\mathsf{Dock} \sqsubseteq \mathsf{LabArtifact}\\
&\mathsf{Dock} \sqsubseteq \exists \mathit{hasInteractionPose}.\mathsf{DockPose}
\end{aligned}$$

DeterministicAMRDock V-guide dock with pose sensors (H2).

$$\begin{aligned}
&\mathsf{DeterministicAMRDock} \sqsubseteq \mathsf{Dock}\\
&\mathsf{DeterministicAMRDock} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{DockSensor}
\end{aligned}$$

DispatchDock Loading / charging dock in the storage room (S1).

$$\begin{aligned}
&\mathsf{DispatchDock} \sqsubseteq \mathsf{Dock}\\
&\mathsf{DispatchDock} \sqsubseteq \exists \mathit{locatedAt}.\mathsf{StorageRoom}
\end{aligned}$$

ToolMagazine Automatic tool magazine (TCM01, Zimmer WMR2063 baseline).

$$\begin{aligned}
&\mathsf{ToolMagazine} \sqsubseteq \mathsf{LabArtifact}\\
&\mathsf{ToolMagazine} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{MagazineStand}\\
&\mathsf{ToolMagazine} \sqsubseteq \exists \mathit{storesTool}.\mathsf{ExchangeableTool}
\end{aligned}$$

PassiveFixture A fixture with no moving joints.

$$\begin{aligned}
&\mathsf{PassiveFixture} \sqsubseteq \mathsf{LabArtifact}\\
&\mathsf{PassiveFixture} \sqsubseteq \neg\exists \mathit{hasDirectComponent}.\mathsf{MechanismJoint}
\end{aligned}$$

HandoverNest Passive nest where two robots hand a plate over (R01/R02).

$$\begin{aligned}
&\mathsf{HandoverNest} \sqsubseteq \mathsf{PassiveFixture}\\
&\mathsf{HandoverNest} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{SeatSensor}\\
&\mathsf{HandoverNest} \sqsubseteq \geq 2\ \mathit{isServedByRobot}.\mathsf{Robot}
\end{aligned}$$

ManualLabEquipment Laboratory equipment operated by people, with no robot control interface.

$$\begin{aligned}
&\mathsf{ManualLabEquipment} \sqsubseteq \mathsf{LabArtifact}\\
&\mathsf{ManualLabEquipment} \sqsubseteq \neg\mathsf{LabAutomationDevice}\\
&\mathsf{ManualLabEquipment} \sqsubseteq \exists \mathit{hasEquipmentFunction}.\mathsf{DeviceFunction}
\end{aligned}$$

BiosafetyCabinet Class II BSC.

$$\begin{aligned}
&\mathsf{BiosafetyCabinet} \sqsubseteq \mathsf{ManualLabEquipment}\\
&\mathsf{BiosafetyCabinet} \sqsubseteq \exists \mathit{hasEquipmentFunction}.\mathsf{AsepticWorkFunction}
\end{aligned}$$

ManualCO2Incubator Benchtop.

$$\begin{aligned}
&\mathsf{ManualCO2Incubator} \sqsubseteq \mathsf{ManualLabEquipment}\\
&\mathsf{ManualCO2Incubator} \sqsubseteq \exists \mathit{hasEquipmentFunction}.\mathsf{IncubationFunction}
\end{aligned}$$

InvertedMicroscope Manual scope.

$$\begin{aligned}
&\mathsf{InvertedMicroscope} \sqsubseteq \mathsf{ManualLabEquipment}\\
&\mathsf{InvertedMicroscope} \sqsubseteq \exists \mathit{hasEquipmentFunction}.\mathsf{ImagingFunction}
\end{aligned}$$

Centrifuge Benchtop centrifuge.

$$\begin{aligned}
&\mathsf{Centrifuge} \sqsubseteq \mathsf{ManualLabEquipment}\\
&\mathsf{Centrifuge} \sqsubseteq \exists \mathit{hasEquipmentFunction}.\mathsf{SeparationFunction}
\end{aligned}$$

WaterBath Water bath.

$$\begin{aligned}
&\mathsf{WaterBath} \sqsubseteq \mathsf{ManualLabEquipment}\\
&\mathsf{WaterBath} \sqsubseteq \exists \mathit{hasEquipmentFunction}.\mathsf{TemperatureControlFunction}
\end{aligned}$$

Autoclave Steam steriliser.

$$\mathsf{Autoclave} \sqsubseteq \mathsf{ManualLabEquipment}$$
$$\mathsf{Autoclave} \sqsubseteq \exists \mathit{hasEquipmentFunction}.\mathsf{SterilizationFunction}$$

ColdStorageDevice Cold storage.

$$\mathsf{ColdStorageDevice} \sqsubseteq \mathsf{ManualLabEquipment}$$
$$\mathsf{ColdStorageDevice} \sqsubseteq \exists \mathit{hasEquipmentFunction}.\mathsf{ColdStorageFunction}$$

LabRefrigerator +4 °C refrigerator.

$$\mathsf{LabRefrigerator} \sqsubseteq \mathsf{ColdStorageDevice}$$
$$\mathsf{LabRefrigerator} \sqsubseteq \exists \mathit{locatedAt}.\mathsf{StorageRoom}$$

LabFreezer -20 °C freezer.

$$\mathsf{LabFreezer} \sqsubseteq \mathsf{ColdStorageDevice}$$
$$\mathsf{LabFreezer} \sqsubseteq \exists \mathit{locatedAt}.\mathsf{StorageRoom}$$

ULTFreezer -80 °C ultra-low freezer.

$$\mathsf{ULTFreezer} \sqsubseteq \mathsf{ColdStorageDevice}$$
$$\mathsf{ULTFreezer} \sqsubseteq \exists \mathit{locatedAt}.\mathsf{StorageRoom}$$

CryogenicDewar LN2 dewar.

$$\mathsf{CryogenicDewar} \sqsubseteq \mathsf{ColdStorageDevice}$$
$$\mathsf{CryogenicDewar} \sqsubseteq \exists \mathit{isMonitoredBy}.\mathsf{O2DepletionMonitor}$$

GasCylinderCabinet Ventilated CO2 cabinet.

$$\mathsf{GasCylinderCabinet} \sqsubseteq \mathsf{SafetyDevice}$$
$$\mathsf{GasCylinderCabinet} \sqsubseteq \exists \mathit{mitigatesHazard}.\mathsf{AsphyxiationHazard}$$

MechanismPart A component of a lab device or fixture (not a robot-body link; D-LAT-1).

$$\mathsf{MechanismPart} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{MechanismPart} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{LabArtifact}$$

MechanismJoint A kinematic joint of a device or tool mechanism (USD physics joint), typed by AICOR's JointType.

$$\mathsf{MechanismJoint} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{MechanismJoint} \sqsubseteq =1\ \mathit{hasMechanismJointType}.\mathsf{JointType}$$

DrivenMechanismJoint Defined: a mechanism joint with an actuator (a USD joint with a drive).

$$\mathsf{DrivenMechanismJoint} \sqsubseteq \mathsf{MechanismJoint}$$
$$\mathsf{DrivenMechanismJoint} \equiv \mathsf{MechanismJoint} \sqcap \exists \mathit{isDrivenBy}.\mathsf{Actuator}$$

Actuator Motor or drive of a joint.

$$\mathsf{Actuator} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{Actuator} \sqsubseteq \exists \mathit{drivesJoint}.\mathsf{MechanismJoint}$$

StationHousing Body of a device.

$$\mathsf{StationHousing} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{StationHousing} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{LabAutomationDevice}$$

IncubationChamber Conditioned, insulated chamber.

$$\mathsf{IncubationChamber} \sqsubseteq \mathsf{StationHousing}$$
$$\mathsf{IncubationChamber} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{IncubationDevice}$$

AccessGate Sliding load gate.

$$\mathsf{AccessGate} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{AccessGate} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{LabAutomationDevice}$$

ServiceDoor Hinged service door.

$$\mathsf{ServiceDoor} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{ServiceDoor} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{LabAutomationDevice}$$

LockableDoor Door that a latch bolt locks (H4).

$$\mathsf{LockableDoor} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{LockableDoor} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{QuarantineStation}$$
$$\mathsf{LockableDoor} \sqsubseteq \exists \mathit{isMonitoredBy}.\mathsf{DeviceSensor}$$

LatchBolt Bolt that locks a door.

$$\mathsf{LatchBolt} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{LatchBolt} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{QuarantineStation}$$

OperatorControl A physical control operated by a person.

$$\mathsf{OperatorControl} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{OperatorControl} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{LabArtifact}$$

KeySwitch Key-operated selector (H4 release).

$$\mathsf{KeySwitch} \sqsubseteq \mathsf{OperatorControl}$$
$$\mathsf{KeySwitch} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{KeyCylinder}$$

KeyCylinder Rotating cylinder of a key switch.

$$\mathsf{KeyCylinder} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{KeyCylinder} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{KeySwitch}$$

PushButton Momentary push button.

$$\mathsf{PushButton} \sqsubseteq \mathsf{OperatorControl}$$
$$\mathsf{PushButton} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{ButtonPlunger}$$

ButtonPlunger Moving plunger of a button.

$$\mathsf{ButtonPlunger} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{ButtonPlunger} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{PushButton}$$

PlateShuttle Sliding SBS nest that presents a plate outside the device.

$$\mathsf{PlateShuttle} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{PlateShuttle} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{SeatSensor}$$
$$\mathsf{PlateShuttle} \sqsubseteq \exists \mathit{hasInteractionPose}.\mathsf{PlateSeatPose}$$

PlateCassette Stack of plate slots.

$$\mathsf{PlateCassette} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{PlateCassette} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{PlateSlot}$$

PlateSlot One plate position of a cassette.

$$\mathsf{PlateSlot} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{PlateSlot} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{PlateCassette}$$
$$\mathsf{PlateSlot} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{SlotOccupancySensor}$$
$$\mathsf{PlateSlot} \sqsubseteq \exists \mathit{slotIndex}.\mathsf{xsd{:}integer}$$

InternalPlateHandler Lift moving plates cassette <-> shuttle.

$$\mathsf{InternalPlateHandler} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{InternalPlateHandler} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{PlateFork}$$

PlateFork Fork tine lifting a plate.

$$\mathsf{PlateFork} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{PlateFork} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{InternalPlateHandler}$$

GantryAxis Linear gantry axis.

$$\mathsf{GantryAxis} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{GantryAxis} \sqsubseteq \exists \mathit{isDirectComponentOf}.(\mathsf{DispensingDevice} \sqcup \mathsf{GantryAxis})$$

PipettingChannel Z axis with a pipetting head.

$$\mathsf{PipettingChannel} \sqsubseteq \mathsf{GantryAxis}$$
$$\mathsf{PipettingChannel} \sqsubseteq \exists \mathit{hasInteractionPose}.\mathsf{ToolCenterPose}$$

MotorizedStage XY stage axis (L17's stage is also the load drawer).

$$\mathsf{MotorizedStage} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{MotorizedStage} \sqsubseteq \exists \mathit{isDirectComponentOf}.(\mathsf{ImagingDevice} \sqcup \mathsf{MotorizedStage})$$

FocusDrive Z focus drive.

$$\mathsf{FocusDrive} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{FocusDrive} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{ObjectiveTurret}$$

ObjectiveTurret Objective changer.

$$\mathsf{ObjectiveTurret} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{ObjectiveTurret} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{FocusDrive}$$

MagazineStand Stand carrying tool docks and their occupancy sensors.

$$\mathsf{MagazineStand} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{MagazineStand} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{ToolMagazine}$$
$$\mathsf{MagazineStand} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{SlotOccupancySensor}$$

LinearRail Linear axis carrying a robot (igus ZLW-1660).

$$\mathsf{LinearRail} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{LinearRail} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{RailCarriage}$$

RailCarriage Moving rail carriage.

$$\mathsf{RailCarriage} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{RailCarriage} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{LinearRail}$$

DeviceSensor A sensor of a device or fixture (not AICOR's robot-mounted Sensor; D-LAT-1).

$$\mathsf{DeviceSensor} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{DeviceSensor} \sqsubseteq \exists \mathit{monitorsEntity}.\mathsf{Entity}$$

SeatSensor Flush seat window (plate seated).

$$\mathsf{SeatSensor} \sqsubseteq \mathsf{DeviceSensor}$$
$$\mathsf{SeatSensor} \sqsubseteq \exists \mathit{monitorsEntity}.\mathsf{PlateSeatPose}$$

SlotOccupancySensor Ray occupancy sensor.

$$\mathsf{SlotOccupancySensor} \sqsubseteq \mathsf{DeviceSensor}$$
$$\mathsf{SlotOccupancySensor} \sqsubseteq \exists \mathit{isDirectComponentOf}.(\mathsf{PlateSlot} \sqcup \mathsf{MagazineStand})$$

CarrierPresenceSensor Carrier present in AMR cabinet.

$$\mathsf{CarrierPresenceSensor} \sqsubseteq \mathsf{DeviceSensor}$$
$$\mathsf{CarrierPresenceSensor} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{MobileRobotBase}$$

ThroughBeamSensor Emitter / receiver beam element.

$$\mathsf{ThroughBeamSensor} \sqsubseteq \mathsf{DeviceSensor}$$
$$\mathsf{ThroughBeamSensor} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{LabAutomationDevice}$$

DockSensor Ray measuring the docked AMR pose.

$$\mathsf{DockSensor} \sqsubseteq \mathsf{DeviceSensor}$$
$$\mathsf{DockSensor} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{DeterministicAMRDock}$$

BarcodeReader Reads plate barcodes.

$$\mathsf{BarcodeReader} \sqsubseteq \mathsf{DeviceSensor}$$
$$\mathsf{BarcodeReader} \sqsubseteq \exists \mathit{monitorsEntity}.\mathsf{Labware}$$

LightBarrierSensor Crossed barrier measuring a TCP.

$$\mathsf{LightBarrierSensor} \sqsubseteq \mathsf{DeviceSensor}$$
$$\mathsf{LightBarrierSensor} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{QCCalibrationStation}$$

FiducialMarker Marker used to register poses.

$$\mathsf{FiducialMarker} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{FiducialMarker} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{CalibratingTask}$$

AprilTagMarker AprilTag 36h11.

$$\mathsf{AprilTagMarker} \sqsubseteq \mathsf{FiducialMarker}$$
$$\mathsf{AprilTagMarker} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{LabArtifact}$$

DatumPin Conical registration pin.

$$\mathsf{DatumPin} \sqsubseteq \mathsf{FiducialMarker}$$
$$\mathsf{DatumPin} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{LabArtifact}$$

Cell robots, robot bodies, tools and robot skills

CellRobot A robot of an automation cell, assigned to stations (layout P1).

$$\mathsf{CellRobot} \sqsubseteq \mathsf{Robot}$$
$$\mathsf{CellRobot} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{CellRobot} \sqsubseteq \exists \mathit{servesStation}.\mathsf{LabArtifact}$$
$$\mathsf{CellRobot} \sqsubseteq \exists \mathit{hasRobotSkill}.\mathsf{RobotSkill}$$

FixedBaseManipulator Arm on a fixed pedestal (R01 FR3).

$$\mathsf{FixedBaseManipulator} \sqsubseteq \mathsf{CellRobot}$$
$$\mathsf{FixedBaseManipulator} \sqsubseteq \exists \mathit{hasProperPart}.\mathsf{ArmRobotBody}$$
$$\mathsf{FixedBaseManipulator} \sqsubseteq \neg\exists \mathit{hasDirectComponent}.(\mathsf{LinearRail} \sqcup \mathsf{MobileRobotBase})$$

RailMountedManipulator Arm on a linear rail with a tool changer (R02 UR5e).

$$\mathsf{RailMountedManipulator} \sqsubseteq \mathsf{CellRobot}$$
$$\mathsf{RailMountedManipulator} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{LinearRail}$$
$$\mathsf{RailMountedManipulator} \sqsubseteq \exists \mathit{hasProperPart}.\mathsf{ArmRobotBody}$$

SampleTransportAMR Clean-sample AMR with a sealed cabinet (R03, MiR100 class).

$$\mathsf{SampleTransportAMR} \sqsubseteq \mathsf{CellRobot}$$
$$\mathsf{SampleTransportAMR} \sqsubseteq \exists \mathit{hasProperPart}.\mathsf{MobileRobotBase}$$
$$\mathsf{SampleTransportAMR} \sqsubseteq \exists \mathit{hasRobotSkill}.\mathsf{SampleTransportSkill}$$

ArmRobotBody Serial arm body (FR3, UR5e).

$$\mathsf{ArmRobotBody} \sqsubseteq \mathsf{RobotBody}$$
$$\mathsf{ArmRobotBody} \sqsubseteq \exists \mathit{hasEndEffectorMount}.\mathsf{EndEffector}$$

MobileRobotBase AMR base with the sample cabinet (R03).

$$\mathsf{MobileRobotBase} \sqsubseteq \mathsf{RobotBody}$$
$$\mathsf{MobileRobotBase} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{CabinetLid}$$

CabinetLid Flat lid over the carrier nest.

$$\mathsf{CabinetLid} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{CabinetLid} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{MobileRobotBase}$$

VirtualPlanarLink A simulation link of the AMR's planar base joints (twin artefact).

$$\mathsf{VirtualPlanarLink} \sqsubseteq \mathsf{Link}$$
$$\mathsf{VirtualPlanarLink} \sqsubseteq \exists \mathit{partOf}.\mathsf{MobileRobotBase}$$

AutomaticToolChanger Robot-side tool coupler.

$$\mathsf{AutomaticToolChanger} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{AutomaticToolChanger} \sqsubseteq \exists \mathit{couplesTool}.\mathsf{ExchangeableTool}$$

ExchangeableTool End effector exchanged by a tool changer (stored in a magazine, not a robot-body link).

$$\mathsf{ExchangeableTool} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{ExchangeableTool} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{ToolBody}$$
$$\mathsf{ExchangeableTool} \sqsubseteq \exists \mathit{hasInteractionPose}.\mathsf{ToolCouplingPose}$$

ToolBody Housing of a tool.

$$\mathsf{ToolBody} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{ToolBody} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{ExchangeableTool}$$

GripperJaw Jaw of a gripper tool.

$$\mathsf{GripperJaw} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{GripperJaw} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{ExchangeableTool}$$

PlateGripperTool Two-jaw SBS plate gripper (T01).

$$\mathsf{PlateGripperTool} \sqsubseteq \mathsf{ExchangeableTool}$$
$$\mathsf{PlateGripperTool} \sqsubseteq =2\ \mathit{hasDirectComponent}.\mathsf{GripperJaw}$$
$$\mathsf{PlateGripperTool} \sqsubseteq \exists \mathit{affordsRobotSkill}.\mathsf{PlateTransferSkill}$$
$$\mathsf{PlateGripperTool} \sqcap \mathsf{TubeGripperTool} \sqsubseteq \bot$$

TubeGripperTool T02.

$$\mathsf{TubeGripperTool} \sqsubseteq \mathsf{ExchangeableTool}$$
$$\mathsf{TubeGripperTool} \sqsubseteq \exists \mathit{affordsRobotSkill}.\mathsf{TubeHandlingSkill}$$
$$\mathsf{TubeGripperTool} \sqcap \mathsf{PlateGripperTool} \sqsubseteq \bot$$

PipettingTool T03.

$$\mathsf{PipettingTool} \sqsubseteq \mathsf{ExchangeableTool}$$
$$\mathsf{PipettingTool} \sqsubseteq \exists \mathit{affordsRobotSkill}.\mathsf{LiquidTransferSkill}$$
$$\mathsf{PipettingTool} \sqcap \mathsf{PlateGripperTool} \sqsubseteq \bot$$

SuctionPickupTool T04.

$$\mathsf{SuctionPickupTool} \sqsubseteq \mathsf{ExchangeableTool}$$
$$\mathsf{SuctionPickupTool} \sqsubseteq \exists \mathit{affordsRobotSkill}.\mathsf{PlateTransferSkill}$$
$$\mathsf{SuctionPickupTool} \sqcap \mathsf{PlateGripperTool} \sqsubseteq \bot$$

CameraInspectionTool T05.

$$\mathsf{CameraInspectionTool} \sqsubseteq \mathsf{ExchangeableTool}$$
$$\mathsf{CameraInspectionTool} \sqsubseteq \exists \mathit{affordsRobotSkill}.\mathsf{InspectionSkill}$$
$$\mathsf{CameraInspectionTool} \sqcap \mathsf{PlateGripperTool} \sqsubseteq \bot$$

RobotSkill A capability of a robot.

$$\mathsf{RobotSkill} \sqsubseteq \mathsf{dul{:}Quality}$$
$$\mathsf{RobotSkill} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{LabTaskType}$$

PlateTransferSkill Pick, carry, place SBS plates.

$$\mathsf{PlateTransferSkill} \sqsubseteq \mathsf{RobotSkill}$$
$$\mathsf{PlateTransferSkill} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{PlateTransferTask}$$

SampleTransportSkill Carry samples in a closed cabinet.

$$\mathsf{SampleTransportSkill} \sqsubseteq \mathsf{RobotSkill}$$
$$\mathsf{SampleTransportSkill} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{PlateTransferTask}$$

NavigationSkill Navigate between docks.

$$\mathsf{NavigationSkill} \sqsubseteq \mathsf{RobotSkill}$$
$$\mathsf{NavigationSkill} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{NavigationTask}$$

DockingSkill Dock deterministically.

$$\mathsf{DockingSkill} \sqsubseteq \mathsf{RobotSkill}$$
$$\mathsf{DockingSkill} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{DockingTask}$$

ToolExchangeSkill Exchange end effectors.

$$\mathsf{ToolExchangeSkill} \sqsubseteq \mathsf{RobotSkill}$$
$$\mathsf{ToolExchangeSkill} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{ToolExchangeTask}$$

TubeHandlingSkill Handle tubes.

$$\mathsf{TubeHandlingSkill} \sqsubseteq \mathsf{RobotSkill}$$
$$\mathsf{TubeHandlingSkill} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{PlateTransferTask}$$

LiquidTransferSkill Pipette.

$$\mathsf{LiquidTransferSkill} \sqsubseteq \mathsf{RobotSkill}$$
$$\mathsf{LiquidTransferSkill} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{LiquidDispensingTask}$$

InspectionSkill Inspect with a camera.

$$\mathsf{InspectionSkill} \sqsubseteq \mathsf{RobotSkill}$$
$$\mathsf{InspectionSkill} \sqsubseteq \exists \mathit{functionAffords}.\mathsf{ImagingTask}$$

Labware, wells and material portions

Labware Containers handled in experiments.

$$\mathsf{Labware} \sqsubseteq \mathsf{LabArtifact}$$
$$\mathsf{Labware} \sqsubseteq \exists \mathit{handledIn}.\mathsf{PlateTransferTask}$$

Microplate Defined: labware with wells.

$$\mathsf{Microplate} \sqsubseteq \mathsf{Labware}$$
$$\mathsf{Microplate} \equiv \mathsf{Labware} \sqcap \exists \mathit{hasDirectComponent}.\mathsf{Well}$$

SBSMicroplate96 SBS / ANSI 96-well plate.

$$\mathsf{SBSMicroplate96} \sqsubseteq \mathsf{Microplate}$$
$$\mathsf{SBSMicroplate96} \sqsubseteq {=}96\ \mathit{hasDirectComponent}.\mathsf{Well}$$

CulturePlate Plate holding cultures (Plate_P17).

$$\mathsf{CulturePlate} \sqsubseteq \mathsf{Microplate}$$
$$\mathsf{CulturePlate} \sqsubseteq \exists \mathit{hasDirectComponent}.\mathsf{CultureWell}$$

ReferencePlate L21 calibration reference plate.

$$\mathsf{ReferencePlate} \sqsubseteq \mathsf{Microplate}$$
$$\mathsf{ReferencePlate} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{CalibratingTask}$$

Well A well of a plate.

$$\mathsf{Well} \sqsubseteq \mathsf{MechanismPart}$$
$$\mathsf{Well} \sqsubseteq \exists \mathit{isDirectComponentOf}.\mathsf{Microplate}$$
$$\mathsf{Well} \sqsubseteq \exists \mathit{wellId}.\mathsf{xsd{:}string}$$
$$\mathsf{Well} \sqcap \mathsf{CultureMediumPortion} \sqsubseteq \bot$$

CultureWell Defined: a well that contains culture medium.

$$\mathsf{CultureWell} \sqsubseteq \mathsf{Well}$$
$$\mathsf{CultureWell} \equiv \mathsf{Well} \sqcap \exists \mathit{containsMaterial}.\mathsf{CultureMediumPortion}$$

CultureMediumPortion The medium in one well: a material portion, contained, not a part.

$$\mathsf{CultureMediumPortion} \sqsubseteq \mathsf{BiologicalSample}$$
$$\mathsf{CultureMediumPortion} \sqsubseteq \exists \mathit{isContainedIn}.\mathsf{Well}$$
$$\mathsf{CultureMediumPortion} \sqcap \mathsf{Well} \sqsubseteq \bot$$

WellOrganoid The organoid cultured in one well (Cell 01: hepatotoxicity model; RS reading: a liver organoid).

$$\mathsf{WellOrganoid} \sqsubseteq \mathsf{BiologicalSpecimen}$$
$$\mathsf{WellOrganoid} \sqsubseteq \exists \mathit{locatedAt}.\mathsf{Well}$$

ReagentPortion A reagent portion in a container.

$$\mathsf{ReagentPortion} \sqsubseteq \mathsf{Sample}$$
$$\mathsf{ReagentPortion} \sqsubseteq \exists \mathit{isContainedIn}.\mathsf{Labware}$$

ChallengeReagentPortion Abstract challenge reagent.

$$\mathsf{ChallengeReagentPortion} \sqsubseteq \mathsf{ReagentPortion}$$
$$\mathsf{ChallengeReagentPortion} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{ChallengeDosingTask}$$
$$\mathsf{ChallengeReagentPortion} \sqcap \mathsf{RescueReagentPortion} \sqsubseteq \bot$$

RescueReagentPortion Abstract rescue reagent.

$$\mathsf{RescueReagentPortion} \sqsubseteq \mathsf{ReagentPortion}$$
$$\mathsf{RescueReagentPortion} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{RescueDosingTask}$$
$$\mathsf{RescueReagentPortion} \sqcap \mathsf{ChallengeReagentPortion} \sqsubseteq \bot$$

Poses: coordinate and interaction frames of the twin (AICOR SpatialPose)

TwinPose A named pose on an object (USD Xform).

$$\mathsf{TwinPose} \sqsubseteq \mathsf{SpatialPose}$$
$$\mathsf{TwinPose} \sqsubseteq \exists \mathit{isPoseOf}.\mathsf{dul{:}PhysicalObject}$$

KinematicPose Reference pose of a link or mechanism.

$$\mathsf{KinematicPose} \sqsubseteq \mathsf{TwinPose}$$
$$\mathsf{KinematicPose} \sqsubseteq \exists \mathit{isPoseOf}.\mathsf{LabArtifact}$$

InteractionPose Pose where a task is performed.

$$\mathsf{InteractionPose} \sqsubseteq \mathsf{TwinPose}$$
$$\mathsf{InteractionPose} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{LabTaskType}$$

PlateSeatPose SBS plate seat.

$$\mathsf{PlateSeatPose} \sqsubseteq \mathsf{InteractionPose}$$
$$\mathsf{PlateSeatPose} \sqsubseteq \exists \mathit{affordsTaskType}.(\mathsf{PlateTransferTask} \sqcup \mathsf{PlateStoringTask})$$
$$\mathsf{PlateSeatPose} \sqcap \mathsf{ApproachPose} \sqsubseteq \bot$$

ApproachPose Pose reached before an interaction.

$$\mathsf{ApproachPose} \sqsubseteq \mathsf{InteractionPose}$$
$$\mathsf{ApproachPose} \sqsubseteq \exists \mathit{affordsTaskType}.(\mathsf{NavigationTask} \sqcup \mathsf{PlateTransferTask})$$
$$\mathsf{ApproachPose} \sqcap \mathsf{PlateSeatPose} \sqsubseteq \bot$$

DockPose Pose of a docked AMR.

$$\mathsf{DockPose} \sqsubseteq \mathsf{InteractionPose}$$
$$\mathsf{DockPose} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{DockingTask}$$
$$\mathsf{DockPose} \sqcap \mathsf{PlateSeatPose} \sqsubseteq \bot$$

ToolCenterPose Tool centre point.

$$\mathsf{ToolCenterPose} \sqsubseteq \mathsf{InteractionPose}$$
$$\mathsf{ToolCenterPose} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{PlateTransferTask}$$
$$\mathsf{ToolCenterPose} \sqcap \mathsf{PlateSeatPose} \sqsubseteq \bot$$

ToolCouplingPose Coupling face of a tool or changer.

$$\mathsf{ToolCouplingPose} \sqsubseteq \mathsf{InteractionPose}$$
$$\mathsf{ToolCouplingPose} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{ToolExchangeTask}$$
$$\mathsf{ToolCouplingPose} \sqcap \mathsf{PlateSeatPose} \sqsubseteq \bot$$

PipettingTargetPose Where a channel doses into a well.

$$\begin{aligned}&\mathsf{PipettingTargetPose} \sqsubseteq \mathsf{InteractionPose}\\&\mathsf{PipettingTargetPose} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{LiquidDispensingTask}\\&\mathsf{PipettingTargetPose} \sqcap \mathsf{PlateSeatPose} \sqsubseteq \bot\end{aligned}$$

ImagingTargetPose Where a well is imaged.

$$\begin{aligned}&\mathsf{ImagingTargetPose} \sqsubseteq \mathsf{InteractionPose}\\&\mathsf{ImagingTargetPose} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{ImagingTask}\\&\mathsf{ImagingTargetPose} \sqcap \mathsf{PlateSeatPose} \sqsubseteq \bot\end{aligned}$$

RackPositionPose Rack position.

$$\begin{aligned}&\mathsf{RackPositionPose} \sqsubseteq \mathsf{InteractionPose}\\&\mathsf{RackPositionPose} \sqsubseteq \exists \mathit{affordsTaskType}.(\mathsf{LiquidDispensingTask} \sqcup \mathsf{TipHandlingTask})\\&\mathsf{RackPositionPose} \sqcap \mathsf{PlateSeatPose} \sqsubseteq \bot\end{aligned}$$

WastePose Tip ejection.

$$\begin{aligned}&\mathsf{WastePose} \sqsubseteq \mathsf{InteractionPose}\\&\mathsf{WastePose} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{TipHandlingTask}\\&\mathsf{WastePose} \sqcap \mathsf{PlateSeatPose} \sqsubseteq \bot\end{aligned}$$

CalibrationPose Datum for QC / calibration.

$$\begin{aligned}&\mathsf{CalibrationPose} \sqsubseteq \mathsf{InteractionPose}\\&\mathsf{CalibrationPose} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{CalibratingTask}\\&\mathsf{CalibrationPose} \sqcap \mathsf{PlateSeatPose} \sqsubseteq \bot\end{aligned}$$

StorageLocationPose Shelf location.

$$\begin{aligned}&\mathsf{StorageLocationPose} \sqsubseteq \mathsf{InteractionPose}\\&\mathsf{StorageLocationPose} \sqsubseteq \exists \mathit{affordsTaskType}.(\mathsf{ConsumableStoringTask} \sqcup \mathsf{ColdStoringTask})\end{aligned}$$

ManipulationPointPose Handle, door, lid, sash, tap, button.

$$\begin{aligned}&\mathsf{ManipulationPointPose} \sqsubseteq \mathsf{InteractionPose}\\&\mathsf{ManipulationPointPose} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{OperatorInteractionTask}\end{aligned}$$

OperatorPositionPose Where a person works.

$$\begin{aligned}&\mathsf{OperatorPositionPose} \sqsubseteq \mathsf{InteractionPose}\\&\mathsf{OperatorPositionPose} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{OperatorInteractionTask}\end{aligned}$$

ItemAccessPose Where a person takes or puts an item.

$$\begin{aligned}&\mathsf{ItemAccessPose} \sqsubseteq \mathsf{InteractionPose}\\&\mathsf{ItemAccessPose} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{PlateTransferTask}\end{aligned}$$

CleaningPointPose Sink basin, eyewash.

$$\begin{aligned}&\mathsf{CleaningPointPose} \sqsubseteq \mathsf{InteractionPose}\\&\mathsf{CleaningPointPose} \sqsubseteq \exists \mathit{affordsTaskType}.\mathsf{CleaningTask}\end{aligned}$$

OpticalPose Objective, camera or focal-point pose.

$$\begin{aligned}&\mathsf{OpticalPose} \sqsubseteq \mathsf{TwinPose}\\&\mathsf{OpticalPose} \sqsubseteq \exists \mathit{isPoseOf}.(\mathsf{ImagingDevice} \sqcup \mathsf{MechanismPart} \sqcup \mathsf{CellRobot})\end{aligned}$$

IlluminationPose Illumination axis.

$$\begin{aligned}&\mathsf{IlluminationPose} \sqsubseteq \mathsf{OpticalPose}\\&\mathsf{IlluminationPose} \sqsubseteq \exists \mathit{isPoseOf}.\mathsf{ImagingDevice}\end{aligned}$$

ViewpointPose Viewpoint of a virtual twin camera.

$$\begin{aligned}&\mathsf{ViewpointPose} \sqsubseteq \mathsf{TwinPose}\\&\mathsf{ViewpointPose} \sqsubseteq \exists \mathit{isPoseOf}.\mathsf{dul{:}PhysicalPlace}\end{aligned}$$

Lab task types and actions

LabTaskType A type of lab work.

$$\mathsf{LabTaskType} \sqsubseteq \mathsf{dul{:}Concept}$$
$$\mathsf{LabTaskType} \sqsubseteq \forall \mathit{classifiesAction}.\mathsf{LabAction}$$

PlateTransferTask Move labware between seats. Every action of this type involves a Labware.

$$\mathsf{PlateTransferTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{PlateTransferTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{Labware}$$
$$\mathsf{PlateTransferTask} \sqcap \mathsf{PlateStoringTask} \sqsubseteq \bot$$

PlateStoringTask Put a plate into a store. Every action of this type involves a PlateStorageDevice.

$$\mathsf{PlateStoringTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{PlateStoringTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{PlateStorageDevice}$$
$$\mathsf{PlateStoringTask} \sqcap \mathsf{PlateTransferTask} \sqsubseteq \bot$$

PlateRetrievingTask Take a plate out of a device. Every action of this type involves a LabArtifact.

$$\mathsf{PlateRetrievingTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{PlateRetrievingTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{LabArtifact}$$
$$\mathsf{PlateRetrievingTask} \sqcap \mathsf{PlateTransferTask} \sqsubseteq \bot$$

IncubatingTask Keep a plate in an incubator. Every action of this type involves a IncubationDevice.

$$\mathsf{IncubatingTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{IncubatingTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{IncubationDevice}$$
$$\mathsf{IncubatingTask} \sqcap \mathsf{PlateTransferTask} \sqsubseteq \bot$$

LiquidDispensingTask Dispense liquid into a well. Every action of this type involves a DispensingDevice.

$$\mathsf{LiquidDispensingTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{LiquidDispensingTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{DispensingDevice}$$
$$\mathsf{LiquidDispensingTask} \sqcap \mathsf{PlateTransferTask} \sqsubseteq \bot$$

ChallengeDosingTask Dose the challenge reagent. Every action of this type involves a DispensingDevice.

$$\mathsf{ChallengeDosingTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{ChallengeDosingTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{DispensingDevice}$$

RescueDosingTask Dose the rescue reagent. Every action of this type involves a DispensingDevice.

$$\mathsf{RescueDosingTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{RescueDosingTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{DispensingDevice}$$

ImagingTask Acquire an image. Every action of this type involves a ImagingDevice.

$$\mathsf{ImagingTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{ImagingTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{ImagingDevice}$$
$$\mathsf{ImagingTask} \sqcap \mathsf{PlateTransferTask} \sqsubseteq \bot$$

CalibratingTask Check / calibrate against a reference. Every action of this type involves a QCCalibrationStation.

$$\mathsf{CalibratingTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{CalibratingTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{QCCalibrationStation}$$
$$\mathsf{CalibratingTask} \sqcap \mathsf{PlateTransferTask} \sqsubseteq \bot$$

QuarantiningTask Contain a plate. Every action of this type involves a QuarantineStation.

$$\mathsf{QuarantiningTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{QuarantiningTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{QuarantineStation}$$
$$\mathsf{QuarantiningTask} \sqcap \mathsf{PlateTransferTask} \sqsubseteq \bot$$

QuarantineReleaseTask Release a plate after authorisation. Every action of this type involves a QuarantineStation.

$$\mathsf{QuarantineReleaseTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{QuarantineReleaseTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{QuarantineStation}$$

IdentifyingTask Read a plate identity. Every action of this type involves a BarcodeReader.

$$\mathsf{IdentifyingTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{IdentifyingTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{BarcodeReader}$$

NavigationTask Drive between docks. Every action of this type involves a SampleTransportAMR.

$$\mathsf{NavigationTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{NavigationTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{SampleTransportAMR}$$

DockingTask Dock an AMR. Every action of this type involves a Dock.

$$\mathsf{DockingTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{DockingTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{Dock}$$

ToolExchangeTask Couple / release a tool. Every action of this type involves a AutomaticToolChanger.

$$\mathsf{ToolExchangeTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{ToolExchangeTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{AutomaticToolChanger}$$

TipHandlingTask Pick up / eject a tip. Every action of this type involves a PipettingChannel.

$$\mathsf{TipHandlingTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{TipHandlingTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{PipettingChannel}$$

ColdStoringTask Store material cold. Every action of this type involves a ColdStorageDevice.

$$\mathsf{ColdStoringTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{ColdStoringTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{ColdStorageDevice}$$

SterilizingTask Sterilise. Every action of this type involves a Autoclave.

$$\mathsf{SterilizingTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{SterilizingTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{Autoclave}$$

AsepticHandlingTask Handle open cultures. Every action of this type involves a BiosafetyCabinet.

$$\mathsf{AsepticHandlingTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{AsepticHandlingTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{BiosafetyCabinet}$$

TemperatureControllingTask Hold a temperature. Every action of this type involves a ManualLabEquipment.

$$\mathsf{TemperatureControllingTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{TemperatureControllingTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{ManualLabEquipment}$$

ConsumableStoringTask Store consumables. Every action of this type involves a StorageShelving.

$$\mathsf{ConsumableStoringTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{ConsumableStoringTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{StorageShelving}$$

CleaningTask Wash. Every action of this type involves a LabSinkUnit.

$$\mathsf{CleaningTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{CleaningTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{LabSinkUnit}$$

OperatorInteractionTask A person operates equipment. Every action of this type involves a LabArtifact.

$$\mathsf{OperatorInteractionTask} \sqsubseteq \mathsf{LabTaskType}$$
$$\mathsf{OperatorInteractionTask} \sqsubseteq \forall \mathit{classifiesAction}.\exists \mathit{hasParticipantEntity}.\mathsf{LabArtifact}$$

LabAction An action of lab work.

$$\mathsf{LabAction} \sqsubseteq \mathsf{soma{:}Action}$$
$$\mathsf{LabAction} \sqsubseteq \exists \mathit{executesTaskType}.\mathsf{LabTaskType}$$

RobotAction Defined: a lab action a robot takes part in.

$$\mathsf{RobotAction} \sqsubseteq \mathsf{LabAction}$$
$$\mathsf{RobotAction} \equiv \mathsf{LabAction} \sqcap \exists \mathit{hasParticipantEntity}.\mathsf{Robot}$$

TransferAction Defined: a lab action of type PlateTransferTask.

$$\mathsf{TransferAction} \sqsubseteq \mathsf{LabAction}$$
$$\mathsf{TransferAction} \equiv \mathsf{LabAction} \sqcap \exists \mathit{executesTaskType}.\mathsf{PlateTransferTask}$$

StoreAction Defined: a lab action of type PlateStoringTask.

$$\mathsf{StoreAction} \sqsubseteq \mathsf{LabAction}$$
$$\mathsf{StoreAction} \equiv \mathsf{LabAction} \sqcap \exists \mathit{executesTaskType}.\mathsf{PlateStoringTask}$$

RetrieveAction Defined: a lab action of type PlateRetrievingTask.

$$\mathsf{RetrieveAction} \sqsubseteq \mathsf{LabAction}$$
$$\mathsf{RetrieveAction} \equiv \mathsf{LabAction} \sqcap \exists \mathit{executesTaskType}.\mathsf{PlateRetrievingTask}$$

IncubateAction Defined: a lab action of type IncubatingTask.

$$\mathsf{IncubateAction} \sqsubseteq \mathsf{LabAction}$$
$$\mathsf{IncubateAction} \equiv \mathsf{LabAction} \sqcap \exists \mathit{executesTaskType}.\mathsf{IncubatingTask}$$

DoseAction Defined: a lab action of type LiquidDispensingTask.

$$\mathsf{DoseAction} \sqsubseteq \mathsf{LabAction}$$
$$\mathsf{DoseAction} \equiv \mathsf{LabAction} \sqcap \exists \mathit{executesTaskType}.\mathsf{LiquidDispensingTask}$$

ImageAction Defined: a lab action of type ImagingTask.

$$\mathsf{ImageAction} \sqsubseteq \mathsf{LabAction}$$
$$\mathsf{ImageAction} \equiv \mathsf{LabAction} \sqcap \exists \mathit{executesTaskType}.\mathsf{ImagingTask}$$

QCAction Defined: a lab action of type CalibratingTask.

$$\mathsf{QCAction} \sqsubseteq \mathsf{LabAction}$$
$$\mathsf{QCAction} \equiv \mathsf{LabAction} \sqcap \exists \mathit{executesTaskType}.\mathsf{CalibratingTask}$$

QuarantineAction Defined: a lab action of type QuarantiningTask.

$$\mathsf{QuarantineAction} \sqsubseteq \mathsf{LabAction}$$
$$\mathsf{QuarantineAction} \equiv \mathsf{LabAction} \sqcap \exists \mathit{executesTaskType}.\mathsf{QuarantiningTask}$$

AMRTransferAction Defined: a transfer by the sample AMR.

$$\mathsf{AMRTransferAction} \sqsubseteq \mathsf{TransferAction}$$
$$\mathsf{AMRTransferAction} \equiv \mathsf{TransferAction} \sqcap \exists \mathit{hasParticipantEntity}.\mathsf{SampleTransportAMR}$$

Device manual knowledge (Cell 01 component contracts) and device process knowledge

DeviceManual Operating knowledge of a device or robot: identity, procedures, setpoints, constraints, monitors, failure modes, recovery and risk controls.

$$\mathsf{DeviceManual} \sqsubseteq \mathsf{dul{:}InformationObject}$$
$$\mathsf{DeviceManual} \sqsubseteq \exists \mathit{manualFor}.\mathsf{LabArtifact}$$
$$\mathsf{DeviceManual} \sqsubseteq \exists \mathit{documentsProcedure}.\mathsf{OperatingProcedure}$$
$$\mathsf{DeviceManual} \sqsubseteq \exists \mathit{sourceDocument}.\mathsf{xsd{:}string}$$

CellComponentContract A Cell 01 SDT component contract (sdt/components/*.yaml): the project's manual-equivalent; no vendor manual is in the repository (restricted source folders are empty).

$$\mathsf{CellComponentContract} \sqsubseteq \mathsf{DeviceManual}$$
$$\mathsf{CellComponentContract} \sqsubseteq \exists \mathit{manufacturerBaseline}.\mathsf{xsd{:}string}$$

OperatingProcedure A documented device operation with preconditions, parameters, effects, monitors, recovery and evidence outputs (Cell 01 action_contracts).

$$\mathsf{OperatingProcedure} \sqsubseteq \mathsf{ActionContract}$$
$$\mathsf{OperatingProcedure} \sqsubseteq \exists \mathit{requiresPrecondition}.\mathsf{Precondition}$$
$$\mathsf{OperatingProcedure} \sqsubseteq \exists \mathit{hasExpectedEffect}.\mathsf{ExpectedEffect}$$
$$\mathsf{OperatingProcedure} \sqsubseteq \exists \mathit{hasRecoveryProcedure}.\mathsf{RecoveryProcedure}$$

ManualItem One item of a manual, verbatim.

$$\mathsf{ManualItem} \sqsubseteq \mathsf{dul{:}Description}$$
$$\mathsf{ManualItem} \sqsubseteq \exists \mathit{statementText}.\mathsf{xsd{:}string}$$

Precondition Must hold before a procedure.

$$\mathsf{Precondition} \sqsubseteq \mathsf{ManualItem}$$
$$\mathsf{Precondition} \sqsubseteq \exists \mathit{isRequiredBy}.\mathsf{OperatingProcedure}$$

IdentityPrecondition Plate / reagent / target identity verified or consistent.

$$\mathsf{IdentityPrecondition} \sqsubseteq \mathsf{Precondition}$$
$$\mathsf{IdentityPrecondition} \sqsubseteq \exists \mathit{isRequiredBy}.\mathsf{OperatingProcedure}$$
$$\mathsf{IdentityPrecondition} \sqcap \mathsf{AvailabilityPrecondition} \sqsubseteq \bot$$

AvailabilityPrecondition Device / slot / station available, reserved or clear.

$$\mathsf{AvailabilityPrecondition} \sqsubseteq \mathsf{Precondition}$$
$$\mathsf{AvailabilityPrecondition} \sqsubseteq \exists \mathit{isRequiredBy}.\mathsf{OperatingProcedure}$$
$$\mathsf{AvailabilityPrecondition} \sqcap \mathsf{IdentityPrecondition} \sqsubseteq \bot$$

EnvironmentPrecondition Environment within its valid envelope.

$$\mathsf{EnvironmentPrecondition} \sqsubseteq \mathsf{Precondition}$$
$$\mathsf{EnvironmentPrecondition} \sqsubseteq \exists \mathit{isRequiredBy}.\mathsf{OperatingProcedure}$$
$$\mathsf{EnvironmentPrecondition} \sqcap \mathsf{IdentityPrecondition} \sqsubseteq \bot$$

CalibrationPrecondition Calibration / tool frame valid.

$$\mathsf{CalibrationPrecondition} \sqsubseteq \mathsf{Precondition}$$
$$\mathsf{CalibrationPrecondition} \sqsubseteq \exists \mathit{isRequiredBy}.\mathsf{OperatingProcedure}$$
$$\mathsf{CalibrationPrecondition} \sqcap \mathsf{IdentityPrecondition} \sqsubseteq \bot$$

SafetyPrecondition A safety or containment state holds (interlocks ready, airspace clear).

$$\mathsf{SafetyPrecondition} \sqsubseteq \mathsf{Precondition}$$
$$\mathsf{SafetyPrecondition} \sqsubseteq \exists \mathit{isRequiredBy}.\mathsf{OperatingProcedure}$$
$$\mathsf{SafetyPrecondition} \sqcap \mathsf{IdentityPrecondition} \sqsubseteq \bot$$

AuthorizationPrecondition An authorisation is present (quarantine release, QA decision).

$$\mathsf{AuthorizationPrecondition} \sqsubseteq \mathsf{Precondition}$$
$$\mathsf{AuthorizationPrecondition} \sqsubseteq \exists \mathit{isRequiredBy}.\mathsf{OperatingProcedure}$$
$$\mathsf{AuthorizationPrecondition} \sqcap \mathsf{IdentityPrecondition} \sqsubseteq \bot$$

PlacementPrecondition An object is seated / present / positioned.

$$\mathsf{PlacementPrecondition} \sqsubseteq \mathsf{Precondition}$$
$$\mathsf{PlacementPrecondition} \sqsubseteq \exists \mathit{isRequiredBy}.\mathsf{OperatingProcedure}$$
$$\mathsf{PlacementPrecondition} \sqcap \mathsf{IdentityPrecondition} \sqsubseteq \bot$$

ExpectedEffect State after a procedure.

$$\mathsf{ExpectedEffect} \sqsubseteq \mathsf{ManualItem}$$
$$\mathsf{ExpectedEffect} \sqsubseteq \exists \mathit{isEffectOf}.\mathsf{OperatingProcedure}$$

EvidenceOutput Evidence a procedure records.

$$\mathsf{EvidenceOutput} \sqsubseteq \mathsf{ManualItem}$$
$$\mathsf{EvidenceOutput} \sqsubseteq \exists \mathit{isEvidenceOf}.\mathsf{OperatingProcedure}$$

OperatingSetpoint Nominal operating state value of a device (e.g. 37 °C).

$$\mathsf{OperatingSetpoint} \sqsubseteq \mathsf{ManualItem}$$
$$\mathsf{OperatingSetpoint} \sqsubseteq \exists \mathit{quantityName}.\mathsf{xsd{:}string}$$

OperatingConstraint A constraint a manual places on operation.

$$\begin{aligned}\mathsf{OperatingConstraint} &\sqsubseteq \mathsf{ManualItem}\\ \mathsf{OperatingConstraint} &\sqsubseteq \mathsf{LabConstraint}\\ \mathsf{OperatingConstraint} &\sqsubseteq \exists \mathit{constrainsDevice}.\mathsf{LabArtifact}\end{aligned}$$

DeviceMonitor A monitored expectation of a device (expectation + observation channel).

$$\begin{aligned}\mathsf{DeviceMonitor} &\sqsubseteq \mathsf{ProcessMonitor}\\ \mathsf{DeviceMonitor} &\sqsubseteq \exists \mathit{statementText}.\mathsf{xsd{:}string}\end{aligned}$$

DeviceFailureMode A fault of a device, with its consequence and recovery (Cell 01 failure_modes).

$$\begin{aligned}\mathsf{DeviceFailureMode} &\sqsubseteq \mathsf{FailureSignature}\\ \mathsf{DeviceFailureMode} &\sqsubseteq \exists \mathit{hasRecoveryProcedure}.\mathsf{RecoveryProcedure}\\ \mathsf{DeviceFailureMode} &\sqsubseteq \exists \mathit{hasFailureConsequence}.\mathsf{LabHazard}\\ \mathsf{DeviceFailureMode} &\sqsubseteq \exists \mathit{statementText}.\mathsf{xsd{:}string}\end{aligned}$$

RecoveryProcedure What to do when a fault occurs.

$$\begin{aligned}\mathsf{RecoveryProcedure} &\sqsubseteq \mathsf{Repair}\\ \mathsf{RecoveryProcedure} &\sqsubseteq \exists \mathit{statementText}.\mathsf{xsd{:}string}\end{aligned}$$

RiskControl A control measure a manual names (alarms, timing limits, interlocks, identity tracking).

$$\begin{aligned}\mathsf{RiskControl} &\sqsubseteq \mathsf{ManualItem}\\ \mathsf{RiskControl} &\sqsubseteq \mathsf{SafetyConstraint}\\ \mathsf{RiskControl} &\sqsubseteq \exists \mathit{mitigatesHazard}.\mathsf{LabHazard}\end{aligned}$$

DeviceProcessMicrotheory Scoped process knowledge of a device: state variables, articulations, environment model.

$$\begin{aligned}\mathsf{DeviceProcessMicrotheory} &\sqsubseteq \mathsf{BodyWorldProcessMicrotheory}\\ \mathsf{DeviceProcessMicrotheory} &\sqsubseteq \exists \mathit{microtheoryOf}.\mathsf{LabArtifact}\end{aligned}$$

TwinControllerCheck A precondition / rule check implemented in a twin device controller (refuses the command when violated).

$$\begin{aligned}\mathsf{TwinControllerCheck} &\sqsubseteq \mathsf{dul{:}Description}\\ \mathsf{TwinControllerCheck} &\sqsubseteq \exists \mathit{exercisedByTest}.\mathsf{TwinVerificationTest}\end{aligned}$$

TwinVerificationTest A twin regression test (scripts/test_*.py) with its pass record.

$$\begin{aligned}\mathsf{TwinVerificationTest} &\sqsubseteq \mathsf{dul{:}InformationObject}\\ \mathsf{TwinVerificationTest} &\sqsubseteq \exists \mathit{verificationTest}.\mathsf{xsd{:}string}\end{aligned}$$

TwinEnforcedPrecondition Defined: a manual precondition the twin enforces.

$$\begin{aligned}\mathsf{TwinEnforcedPrecondition} &\sqsubseteq \mathsf{Precondition}\\ \mathsf{TwinEnforcedPrecondition} &\equiv \mathsf{Precondition} \sqcap \exists \mathit{enforcedInTwinBy}.\mathsf{TwinControllerCheck}\end{aligned}$$

Safe action: hazards, safety rules, enforcement, admissibility of candidate actions

LabHazard A potential source of harm in the lab.

$$\begin{aligned}\mathsf{LabHazard} &\sqsubseteq \mathsf{dul{:}Description}\\ \mathsf{LabHazard} &\sqsubseteq \exists \mathit{hazardTo}.\mathsf{Entity}\end{aligned}$$

HumanInjuryHazard Robot motion or pinch points near a person.

$$\begin{aligned}\mathsf{HumanInjuryHazard} &\sqsubseteq \mathsf{LabHazard}\\ \mathsf{HumanInjuryHazard} &\sqsubseteq \exists \mathit{hazardTo}.\mathsf{Entity}\end{aligned}$$

CollisionHazard Robot / device / labware collision.

$$\begin{aligned}\mathsf{CollisionHazard} &\sqsubseteq \mathsf{LabHazard}\\ \mathsf{CollisionHazard} &\sqsubseteq \exists \mathit{hazardTo}.\mathsf{Entity}\end{aligned}$$

ContaminationHazard Transfer of contaminants to cultures or reagents.

$$\mathsf{ContaminationHazard} \sqsubseteq \mathsf{LabHazard}$$
$$\mathsf{ContaminationHazard} \sqsubseteq \exists \mathit{hazardTo}.\mathsf{Entity}$$

SampleIntegrityHazard Loss of environment, time window or physical integrity.

$$\mathsf{SampleIntegrityHazard} \sqsubseteq \mathsf{LabHazard}$$
$$\mathsf{SampleIntegrityHazard} \sqsubseteq \exists \mathit{hazardTo}.\mathsf{Entity}$$

IdentityHazard Swap or loss of sample identity.

$$\mathsf{IdentityHazard} \sqsubseteq \mathsf{LabHazard}$$
$$\mathsf{IdentityHazard} \sqsubseteq \exists \mathit{hazardTo}.\mathsf{Entity}$$

MeasurementValidityHazard Measurements made with an invalid measurement chain.

$$\mathsf{MeasurementValidityHazard} \sqsubseteq \mathsf{LabHazard}$$
$$\mathsf{MeasurementValidityHazard} \sqsubseteq \exists \mathit{hazardTo}.\mathsf{Entity}$$

AsphyxiationHazard Oxygen depletion from LN2 / CO2.

$$\mathsf{AsphyxiationHazard} \sqsubseteq \mathsf{LabHazard}$$
$$\mathsf{AsphyxiationHazard} \sqsubseteq \exists \mathit{hazardTo}.\mathsf{Entity}$$

FireHazard Fire.

$$\mathsf{FireHazard} \sqsubseteq \mathsf{LabHazard}$$
$$\mathsf{FireHazard} \sqsubseteq \exists \mathit{hazardTo}.\mathsf{Entity}$$

ChemicalExposureHazard Splash / exposure of a person.

$$\mathsf{ChemicalExposureHazard} \sqsubseteq \mathsf{LabHazard}$$
$$\mathsf{ChemicalExposureHazard} \sqsubseteq \exists \mathit{hazardTo}.\mathsf{Entity}$$

LabSafetyRule A universal prohibition on lab actions in a situation (AICOR IncapabilityAssertion: removes trajectories).

$$\mathsf{LabSafetyRule} \sqsubseteq \mathsf{IncapabilityAssertion}$$
$$\mathsf{LabSafetyRule} \sqsubseteq \exists \mathit{mitigatesHazard}.\mathsf{LabHazard}$$
$$\mathsf{LabSafetyRule} \sqsubseteq \exists \mathit{prohibitsTaskType}.\mathsf{LabTaskType}$$
$$\mathsf{LabSafetyRule} \sqsubseteq \exists \mathit{derivedFromSource}.\mathsf{dul{:}InformationObject}$$
$$\mathsf{LabSafetyRule} \sqsubseteq \exists \mathit{statementText}.\mathsf{xsd{:}string}$$

HumanSafetyRule Protects people (enclosure, door interlocks, light curtain, e-stop).

$$\mathsf{HumanSafetyRule} \sqsubseteq \mathsf{LabSafetyRule}$$
$$\mathsf{HumanSafetyRule} \sqsubseteq \exists \mathit{mitigatesHazard}.\mathsf{HumanInjuryHazard}$$
$$\mathsf{HumanSafetyRule} \sqsubseteq \exists \mathit{enforcedBy}.\mathsf{SafetyDevice}$$
$$\mathsf{HumanSafetyRule} \sqcap \mathsf{MachineProtectionRule} \sqsubseteq \bot$$

MachineProtectionRule Protects equipment and labware (airspace interlock, gate state, cabinet closed in transit).

$$\mathsf{MachineProtectionRule} \sqsubseteq \mathsf{LabSafetyRule}$$
$$\mathsf{MachineProtectionRule} \sqsubseteq \exists \mathit{mitigatesHazard}.\mathsf{CollisionHazard}$$
$$\mathsf{MachineProtectionRule} \sqcap \mathsf{HumanSafetyRule} \sqsubseteq \bot$$

BiosafetyRule Asepsis / contamination rules; gates intervention admissibility (RS ContaminationCompatibility).

$$\mathsf{BiosafetyRule} \sqsubseteq \mathsf{LabSafetyRule}$$
$$\mathsf{BiosafetyRule} \sqsubseteq \mathsf{ContaminationCompatibility}$$
$$\mathsf{BiosafetyRule} \sqsubseteq \exists \mathit{mitigatesHazard}.\mathsf{ContaminationHazard}$$
$$\mathsf{BiosafetyRule} \sqcap \mathsf{HumanSafetyRule} \sqsubseteq \bot$$

SampleIntegrityRule Environment envelope, time outside the incubator, seating.

$$\mathsf{SampleIntegrityRule} \sqsubseteq \mathsf{LabSafetyRule}$$
$$\mathsf{SampleIntegrityRule} \sqsubseteq \exists \mathit{mitigatesHazard}.\mathsf{SampleIntegrityHazard}$$
$$\mathsf{SampleIntegrityRule} \sqcap \mathsf{HumanSafetyRule} \sqsubseteq \bot$$

IdentityRule Identity must be verified.

$$\begin{aligned}&\mathsf{IdentityRule} \sqsubseteq \mathsf{LabSafetyRule}\\&\mathsf{IdentityRule} \sqsubseteq \exists \mathit{mitigatesHazard}.\mathsf{IdentityHazard}\\&\mathsf{IdentityRule} \sqcap \mathsf{HumanSafetyRule} \sqsubseteq \bot\end{aligned}$$

MeasurementValidityRule No measurement with an invalid calibration / tool frame.

$$\begin{aligned}&\mathsf{MeasurementValidityRule} \sqsubseteq \mathsf{LabSafetyRule}\\&\mathsf{MeasurementValidityRule} \sqsubseteq \exists \mathit{mitigatesHazard}.\mathsf{MeasurementValidityHazard}\\&\mathsf{MeasurementValidityRule} \sqcap \mathsf{HumanSafetyRule} \sqsubseteq \bot\end{aligned}$$

AuthorizationNorm A normative rule: an action is permitted only with an authorisation (normative, not causal, admissibility).

$$\begin{aligned}&\mathsf{AuthorizationNorm} \sqsubseteq \mathsf{Norm}\\&\mathsf{AuthorizationNorm} \sqsubseteq \exists \mathit{prohibitsTaskType}.\mathsf{LabTaskType}\\&\mathsf{AuthorizationNorm} \sqsubseteq \exists \mathit{statementText}.\mathsf{xsd{:}string}\end{aligned}$$

SafetySourceDocument A document safety knowledge is derived from (Cell 01 specification, rule files, site proposal).

$$\begin{aligned}&\mathsf{SafetySourceDocument} \sqsubseteq \mathsf{dul{:}InformationObject}\\&\mathsf{SafetySourceDocument} \sqsubseteq \exists \mathit{sourceDocument}.\mathsf{xsd{:}string}\end{aligned}$$

SafetyEnforcementMechanism How a rule is enforced in the twin.

$$\begin{aligned}&\mathsf{SafetyEnforcementMechanism} \sqsubseteq \mathsf{dul{:}Description}\\&\mathsf{SafetyEnforcementMechanism} \sqsubseteq \exists \mathit{enforces}.\mathsf{LabSafetyRule}\end{aligned}$$

PhysicalInterlockMechanism Enforcement by a safety device (door switch, light curtain, e-stop).

$$\begin{aligned}&\mathsf{PhysicalInterlockMechanism} \sqsubseteq \mathsf{SafetyEnforcementMechanism}\\&\mathsf{PhysicalInterlockMechanism} \sqsubseteq \exists \mathit{realizedByDevice}.\mathsf{SafetyDevice}\end{aligned}$$

ControllerInterlockMechanism Enforcement by a device controller refusing a command (L12 airspace, L10 environment, H4 token).

$$\begin{aligned}&\mathsf{ControllerInterlockMechanism} \sqsubseteq \mathsf{SafetyEnforcementMechanism}\\&\mathsf{ControllerInterlockMechanism} \sqsubseteq \mathsf{TwinControllerCheck}\\&\mathsf{ControllerInterlockMechanism} \sqsubseteq \exists \mathit{realizedByDevice}.\mathsf{LabArtifact}\end{aligned}$$

CriticGateMechanism Enforcement by Cell 01's critic vector: an event commits only if safety_ok (AICOR commitment checking).

$$\begin{aligned}&\mathsf{CriticGateMechanism} \sqsubseteq \mathsf{SafetyEnforcementMechanism}\\&\mathsf{CriticGateMechanism} \sqsubseteq \mathsf{SafetyCheck}\\&\mathsf{CriticGateMechanism} \sqsubseteq \exists \mathit{testsCriterion}.\mathsf{CommitmentCriterion}\end{aligned}$$

UnsafeCandidateAction Defined: a candidate action that violates a lab safety rule (inadmissible).

$$\begin{aligned}&\mathsf{UnsafeCandidateAction} \sqsubseteq \mathsf{CandidateAction}\\&\mathsf{UnsafeCandidateAction} \equiv \mathsf{CandidateAction} \sqcap \exists \mathit{violatesRule}.\mathsf{LabSafetyRule}\end{aligned}$$

UnauthorizedCandidateAction Defined: a candidate action that violates an authorisation norm (normatively inadmissible).

$$\begin{aligned}&\mathsf{UnauthorizedCandidateAction} \sqsubseteq \mathsf{CandidateAction}\\&\mathsf{UnauthorizedCandidateAction} \equiv \mathsf{CandidateAction} \sqcap \exists \mathit{violatesRule}.\mathsf{AuthorizationNorm}\end{aligned}$$

RefusedCommand A command the twin refused, observed in a verification test.

$$\begin{aligned}&\mathsf{RefusedCommand} \sqsubseteq \mathsf{CandidateAction}\\&\mathsf{RefusedCommand} \sqsubseteq \exists \mathit{violatesRule}.(\mathsf{LabSafetyRule} \sqcup \mathsf{AuthorizationNorm})\\&\mathsf{RefusedCommand} \sqsubseteq \exists \mathit{refusedByEntity}.\mathsf{Entity}\\&\mathsf{RefusedCommand} \sqsubseteq \exists \mathit{verificationTest}.\mathsf{xsd{:}string}\end{aligned}$$

ProtectiveStopEvent A latched protective stop (e.g. light curtain broken without muting).

$$\begin{aligned}&\mathsf{ProtectiveStopEvent} \sqsubseteq \mathsf{dul{:}Event}\\&\mathsf{ProtectiveStopEvent} \sqsubseteq \exists \mathit{triggeredByDevice}.\mathsf{SafetyDevice}\end{aligned}$$

Culture experiments: designs, arms, interventions, readouts, runs, conclusions

ChallengeRescueStudyDesign Cell 01's challenge-rescue design: graded challenge doses, rescue at 6 h, untreated controls, 4x imaging every 6 h to 72 h.

$$\begin{aligned}
&\mathsf{ChallengeRescueStudyDesign} \sqsubseteq \mathsf{StudyDesign}\\
&\mathsf{ChallengeRescueStudyDesign} \sqsubseteq \exists \mathit{hasArm}.\mathsf{ChallengeOnlyArm}\\
&\mathsf{ChallengeRescueStudyDesign} \sqsubseteq \exists \mathit{hasArm}.\mathsf{ChallengeRescueArm}\\
&\mathsf{ChallengeRescueStudyDesign} \sqsubseteq \exists \mathit{hasArm}.\mathsf{ControlArm}\\
&\mathsf{ChallengeRescueStudyDesign} \sqsubseteq \exists \mathit{observedBySchedule}.\mathsf{ObservationSchedule}
\end{aligned}$$

ExperimentalArm A group of units treated alike.

$$\begin{aligned}
&\mathsf{ExperimentalArm} \sqsubseteq \mathsf{dul{:}Description}\\
&\mathsf{ExperimentalArm} \sqsubseteq \exists \mathit{assignsUnit}.\mathsf{Well}
\end{aligned}$$

InterventionSpec A planned intervention (agent, dose, time).

$$\begin{aligned}
&\mathsf{InterventionSpec} \sqsubseteq \mathsf{dul{:}Description}\\
&\mathsf{InterventionSpec} \sqsubseteq \exists \mathit{doseAU}.\mathsf{xsd{:}double}\\
&\mathsf{InterventionSpec} \sqsubseteq \exists \mathit{time_h}.\mathsf{xsd{:}double}
\end{aligned}$$

ChallengeSpec Planned challenge dose.

$$\begin{aligned}
&\mathsf{ChallengeSpec} \sqsubseteq \mathsf{InterventionSpec}\\
&\mathsf{ChallengeSpec} \sqsubseteq \exists \mathit{specifiesTask}.\mathsf{ChallengeDosingTask}\\
&\mathsf{ChallengeSpec} \sqcap \mathsf{RescueSpec} \sqsubseteq \bot
\end{aligned}$$

RescueSpec Planned rescue dose.

$$\begin{aligned}
&\mathsf{RescueSpec} \sqsubseteq \mathsf{InterventionSpec}\\
&\mathsf{RescueSpec} \sqsubseteq \exists \mathit{specifiesTask}.\mathsf{RescueDosingTask}\\
&\mathsf{RescueSpec} \sqcap \mathsf{ChallengeSpec} \sqsubseteq \bot
\end{aligned}$$

ChallengeOnlyArm Defined: challenged, not rescued.

$$\begin{aligned}
&\mathsf{ChallengeOnlyArm} \sqsubseteq \mathsf{ExperimentalArm}\\
&\mathsf{ChallengeOnlyArm} \equiv \mathsf{ExperimentalArm} \sqcap \exists \mathit{appliesIntervention}.\mathsf{ChallengeSpec} \sqcap \neg\exists \mathit{appliesIntervention}.\mathsf{RescueSpec}\\
&\mathsf{ChallengeOnlyArm} \sqcap \mathsf{ChallengeRescueArm} \sqsubseteq \bot
\end{aligned}$$

ChallengeRescueArm Defined: challenged and rescued.

$$\begin{aligned}
&\mathsf{ChallengeRescueArm} \sqsubseteq \mathsf{ExperimentalArm}\\
&\mathsf{ChallengeRescueArm} \equiv \mathsf{ExperimentalArm} \sqcap \exists \mathit{appliesIntervention}.\mathsf{ChallengeSpec} \sqcap \exists \mathit{appliesIntervention}.\mathsf{RescueSpec}\\
&\mathsf{ChallengeRescueArm} \sqcap \mathsf{ChallengeOnlyArm} \sqsubseteq \bot
\end{aligned}$$

ControlArm Defined: no intervention.

$$\begin{aligned}
&\mathsf{ControlArm} \sqsubseteq \mathsf{ExperimentalArm}\\
&\mathsf{ControlArm} \equiv \mathsf{ExperimentalArm} \sqcap \neg\exists \mathit{appliesIntervention}.\mathsf{InterventionSpec}\\
&\mathsf{ControlArm} \sqcap \mathsf{ChallengeOnlyArm} \sqsubseteq \bot
\end{aligned}$$

ObservationSchedule When units are observed.

$$\begin{aligned}
&\mathsf{ObservationSchedule} \sqsubseteq \mathsf{dul{:}Description}\\
&\mathsf{ObservationSchedule} \sqsubseteq \exists \mathit{interval_h}.\mathsf{xsd{:}double}\\
&\mathsf{ObservationSchedule} \sqsubseteq \exists \mathit{horizon_h}.\mathsf{xsd{:}double}
\end{aligned}$$

ChallengeIntervention A dose of the challenge agent into a well (L12 dose).

$$\begin{aligned}
&\mathsf{ChallengeIntervention} \sqsubseteq \mathsf{BiologicalIntervention}\\
&\mathsf{ChallengeIntervention} \sqsubseteq \exists \mathit{realizedByAction}.\mathsf{ScientificActionDesignator}\\
&\mathsf{ChallengeIntervention} \sqcap \mathsf{RescueIntervention} \sqsubseteq \bot
\end{aligned}$$

RescueIntervention A dose of the rescue agent into a well.

$$\begin{aligned}
&\mathsf{RescueIntervention} \sqsubseteq \mathsf{BiologicalIntervention}\\
&\mathsf{RescueIntervention} \sqsubseteq \exists \mathit{realizedByAction}.\mathsf{ScientificActionDesignator}\\
&\mathsf{RescueIntervention} \sqcap \mathsf{ChallengeIntervention} \sqsubseteq \bot
\end{aligned}$$

MorphologyViabilityReadout Viability inferred from bright-field morphology (radius, core OD, debris, haze) of a 4x image.

$$\mathsf{MorphologyViabilityReadout} \sqsubseteq \mathsf{CellViabilityReadout}$$
$$\mathsf{MorphologyViabilityReadout} \sqsubseteq \exists \mathit{observationOf}.\mathsf{WellOrganoid}$$

RescueHypothesis 'Rescue at 6 h raises 72 h viability at challenge dose d.'

$$\mathsf{RescueHypothesis} \sqsubseteq \mathsf{ScientificHypothesis}$$
$$\mathsf{RescueHypothesis} \sqsubseteq \exists \mathit{heldIn}.\mathsf{CultureSDT}$$

CultureExperimentRun One executed run of the study in the twin.

$$\mathsf{CultureExperimentRun} \sqsubseteq \mathsf{ExperimentalEpisode}$$
$$\mathsf{CultureExperimentRun} \sqsubseteq \exists \mathit{runOfStudy}.\mathsf{ChallengeRescueStudyDesign}$$
$$\mathsf{CultureExperimentRun} \sqsubseteq \exists \mathit{scenarioName}.\mathsf{xsd{:}string}$$

ScenarioCondition A condition of a run (nominal, injected fault).

$$\mathsf{ScenarioCondition} \sqsubseteq \mathsf{dul{:}Description}$$
$$\mathsf{ScenarioCondition} \sqsubseteq \exists \mathit{statementText}.\mathsf{xsd{:}string}$$

FaultInjection A deliberately injected fault.

$$\mathsf{FaultInjection} \sqsubseteq \mathsf{ScenarioCondition}$$
$$\mathsf{FaultInjection} \sqsubseteq \exists \mathit{injectsHazard}.\mathsf{LabHazard}$$

RescueComparisonConclusion Conclusion comparing a rescued with an unrescued well.

$$\mathsf{RescueComparisonConclusion} \sqsubseteq \mathsf{ScientificConclusion}$$
$$\mathsf{RescueComparisonConclusion} \sqsubseteq \geq 2\ \mathit{concernsWell}.\mathsf{Well}$$
$$\mathsf{RescueComparisonConclusion} \sqsubseteq \exists \mathit{claimStatus}.\mathsf{xsd{:}string}$$

WithholdingReason Why a conclusion is withheld.

$$\mathsf{WithholdingReason} \sqsubseteq \mathsf{dul{:}Description}$$
$$\mathsf{WithholdingReason} \sqsubseteq \exists \mathit{statementText}.\mathsf{xsd{:}string}$$

UnverifiedComparatorReason A compared well received an uncommitted (unverified) intervention.

$$\mathsf{UnverifiedComparatorReason} \sqsubseteq \mathsf{WithholdingReason}$$
$$\mathsf{UnverifiedComparatorReason} \sqsubseteq \exists \mathit{citesRule}.\mathsf{LabSafetyRule}$$

ContaminationSuspectedReason A compared well is flagged contamination_suspected.

$$\mathsf{ContaminationSuspectedReason} \sqsubseteq \mathsf{WithholdingReason}$$
$$\mathsf{ContaminationSuspectedReason} \sqsubseteq \exists \mathit{citesRule}.\mathsf{BiosafetyRule}$$

WithheldConclusion Defined: a conclusion withheld for a stated reason.

$$\mathsf{WithheldConclusion} \sqsubseteq \mathsf{RescueComparisonConclusion}$$
$$\mathsf{WithheldConclusion} \equiv \mathsf{RescueComparisonConclusion} \sqcap \exists \mathit{withheldBecause}.\mathsf{WithholdingReason}$$

WellBeliefEstimate The culture SDT's posterior estimate of a well's latent state (epistemic status: inferred).

$$\mathsf{WellBeliefEstimate} \sqsubseteq \mathsf{StateEstimate}$$
$$\mathsf{WellBeliefEstimate} \sqsubseteq \exists \mathit{concernsWell}.\mathsf{Well}$$

WellObservationRecord Measured morphology of one well at one session, with image / calibration provenance.

$$\mathsf{WellObservationRecord} \sqsubseteq \mathsf{CellViabilityReadout}$$
$$\mathsf{WellObservationRecord} \sqsubseteq \exists \mathit{concernsWell}.\mathsf{Well}$$
$$\mathsf{WellObservationRecord} \sqsubseteq \exists \mathit{radius_um}.\mathsf{xsd{:}double}$$

InterventionRecord A dose event in a well's intervention history.

$$\mathsf{InterventionRecord} \sqsubseteq \mathsf{dul{:}InformationObject}$$
$$\mathsf{InterventionRecord} \sqsubseteq \exists \mathit{partOf}.\mathsf{InterventionHistory}$$
$$\mathsf{InterventionRecord} \sqsubseteq \exists \mathit{realizesIntervention}.\mathsf{BiologicalIntervention}$$
$$\mathsf{InterventionRecord} \sqsubseteq \exists \mathit{isCommittedEvent}.\mathsf{xsd{:}boolean}$$

UnverifiedInterventionRecord Defined: an intervention record the ledger did not commit.

$$\mathsf{UnverifiedInterventionRecord} \sqsubseteq \mathsf{InterventionRecord}$$
$$\mathsf{UnverifiedInterventionRecord} \equiv \mathsf{InterventionRecord} \sqcap \exists \mathit{isCommittedEvent}.\mathit{false}$$

WellQualityRecord ESS and anomaly flags.

$$\mathsf{WellQualityRecord} \sqsubseteq \mathsf{dul{:}InformationObject}$$
$$\mathsf{WellQualityRecord} \sqsubseteq \exists \mathit{concernsWell}.\mathsf{Well}$$

ContaminationSuspectedFlag Cell 01's contamination_suspected rule fired for a well.

$$\mathsf{ContaminationSuspectedFlag} \sqsubseteq \mathsf{WellQualityRecord}$$
$$\mathsf{ContaminationSuspectedFlag} \sqsubseteq \exists \mathit{flagsHazard}.\mathsf{ContaminationHazard}$$

Action records, monitor results and routing (Cell 01 NEEM records)

CellActionRecord A Cell 01 NEEM-style record of one action (critic vector, monitors, routing).

$$\mathsf{CellActionRecord} \sqsubseteq \mathsf{LabExecutionTrace}$$
$$\mathsf{CellActionRecord} \sqsubseteq \exists \mathit{recordsAction}.\mathsf{LabAction}$$
$$\mathsf{CellActionRecord} \sqsubseteq \exists \mathit{hasRouting}.\mathsf{RecordRouting}$$

MonitorResult Result of one Cell 01 monitor on an action.

$$\mathsf{MonitorResult} \sqsubseteq \mathsf{dul{:}InformationObject}$$
$$\mathsf{MonitorResult} \sqsubseteq \exists \mathit{monitorPassed}.\mathsf{xsd{:}boolean}$$
$$\mathsf{MonitorResult} \sqsubseteq \exists \mathit{monitorName}.\mathsf{xsd{:}string}$$

FailedMonitorResult Defined: a monitor that did not pass.

$$\mathsf{FailedMonitorResult} \sqsubseteq \mathsf{MonitorResult}$$
$$\mathsf{FailedMonitorResult} \equiv \mathsf{MonitorResult} \sqcap \exists \mathit{monitorPassed}.\mathit{false}$$

BlockedActionRecord Defined: a record with a failed monitor; its semantic event cannot commit.

$$\mathsf{BlockedActionRecord} \sqsubseteq \mathsf{CellActionRecord}$$
$$\mathsf{BlockedActionRecord} \equiv \mathsf{CellActionRecord} \sqcap \exists \mathit{hasMonitorResult}.\mathsf{FailedMonitorResult}$$

RecordRouting A routing outcome.

$$\mathsf{RecordRouting} \sqsubseteq \mathsf{dul{:}Concept}$$
$$\mathsf{RecordRouting} \sqsubseteq \forall \mathit{classifiesRecord}.\mathsf{CellActionRecord}$$

CommitRouting Commit (culture SDT / evidence layer / admission).

$$\mathsf{CommitRouting} \sqsubseteq \mathsf{RecordRouting}$$
$$\mathsf{CommitRouting} \sqsubseteq \exists \mathit{routesTo}.\mathsf{LabSemanticDigitalTwin}$$
$$\mathsf{CommitRouting} \sqcap \mathsf{QuarantineRouting} \sqsubseteq \bot$$

QuarantineRouting Object to H4 (a Repair).

$$\mathsf{QuarantineRouting} \sqsubseteq \mathsf{RecordRouting}$$
$$\mathsf{QuarantineRouting} \sqsubseteq \exists \mathit{routesTo}.\mathsf{QuarantineStation}$$
$$\mathsf{QuarantineRouting} \sqcap \mathsf{CommitRouting} \sqsubseteq \bot$$

EscalationRouting To a human (an AICOR HelpRequest).

$$\mathsf{EscalationRouting} \sqsubseteq \mathsf{RecordRouting}$$
$$\mathsf{EscalationRouting} \sqsubseteq \exists \mathit{routesTo}.\mathsf{dul{:}Agent}$$
$$\mathsf{EscalationRouting} \sqcap \mathsf{CommitRouting} \sqsubseteq \bot$$

InvalidationRouting Invalidate dependent measurements.

$$\mathsf{InvalidationRouting} \sqsubseteq \mathsf{RecordRouting}$$
$$\mathsf{InvalidationRouting} \sqsubseteq \exists \mathit{routesTo}.\mathsf{LabSemanticDigitalTwin}$$
$$\mathsf{InvalidationRouting} \sqcap \mathsf{CommitRouting} \sqsubseteq \bot$$

From a required information gain to the robot's physical actions and motions (derivation layer)

CompetingRecoveryModel One of Cell 01's competing recovery models (mathematical-models.md §5), calibrated to the observed data.

CompetingRecoveryModel $\sqsubseteq$ ScientificModel
CompetingRecoveryModel $\sqsubseteq \exists predictsReadout$.MorphologyViabilityReadout

GradedRecoveryModel M1: P_rec = (0 − 1 I − 2 Δt + 3 d_r); no point of no return.

GradedRecoveryModel $\sqsubseteq$ CompetingRecoveryModel
GradedRecoveryModel $\sqsubseteq \exists statementText$.xsd:string
GradedRecoveryModel $\sqcap$ ThresholdRecoveryModel $\sqsubseteq \bot$

ThresholdRecoveryModel M2: rescue fails once the cumulative injury burden B(Δt) ≥ B*.

ThresholdRecoveryModel $\sqsubseteq$ CompetingRecoveryModel
ThresholdRecoveryModel $\sqsubseteq \exists statementText$.xsd:string
ThresholdRecoveryModel $\sqcap$ GradedRecoveryModel $\sqsubseteq \bot$

ProtectiveThresholdModel M3: B*(t) = B0* + P(t): the threshold moves with adaptive protection.

ProtectiveThresholdModel $\sqsubseteq$ CompetingRecoveryModel
ProtectiveThresholdModel $\sqsubseteq \exists statementText$.xsd:string
ProtectiveThresholdModel $\sqcap$ GradedRecoveryModel $\sqsubseteq \bot$

InformationGainRequirement What an experiment must teach: a minimum expected information gain over the competing models.

InformationGainRequirement $\sqsubseteq$ dul:Description
InformationGainRequirement $\sqsubseteq \exists requiredGain_bits$.xsd:double

ExperimentCandidate A candidate delayed-rescue experiment (dose, rescue delay, replicates) with its EIG and cost.

ExperimentCandidate $\sqsubseteq$ ScientificProposal
ExperimentCandidate $\sqsubseteq \geq 2\ discriminatesModel$.CompetingRecoveryModel
ExperimentCandidate $\sqsubseteq \exists expectedGain_bits$.xsd:double
ExperimentCandidate $\sqsubseteq \exists rescueDelay_h$.xsd:double
ExperimentCandidate $\sqsubseteq \exists doseAU$.xsd:double
ExperimentCandidate $\sqsubseteq \exists replicateCount$.xsd:integer
ExperimentCandidate $\sqsubseteq \exists planCost$.xsd:integer

InformationGainEstimate EIG(a) = E_y KL(p(M|D,y,a) ‖ p(M|D)) (Cell 01 §10), by nested Monte Carlo over the calibrated models.

InformationGainEstimate $\sqsubseteq$ dul:InformationObject
InformationGainEstimate $\sqsubseteq \exists estimatesGainOf$.ExperimentCandidate
InformationGainEstimate $\sqsubseteq \exists conditionedOnBelief$.CultureSDT
InformationGainEstimate $\sqsubseteq \exists expectedGain_bits$.xsd:double

SelectedExperiment Defined: the candidate chosen under a gain requirement (least cost among those meeting it).

SelectedExperiment $\sqsubseteq$ ExperimentCandidate
SelectedExperiment $\equiv$ ExperimentCandidate $\sqcap \exists selectedUnder$.InformationGainRequirement

DelayedRescueStudyDesign A study design with a chosen rescue delay, comparator and untreated control.

DelayedRescueStudyDesign $\sqsubseteq$ StudyDesign
DelayedRescueStudyDesign $\sqsubseteq \exists hasArm$.ChallengeRescueArm
DelayedRescueStudyDesign $\sqsubseteq \exists hasArm$.ChallengeOnlyArm
DelayedRescueStudyDesign $\sqsubseteq \exists hasArm$.ControlArm
DelayedRescueStudyDesign $\sqsubseteq \exists rescueDelay_h$.xsd:double

ExperimentRealisationPlan The derived protocol that realises a selected experiment in the lab: sessions of material, scene and motion designators.

ExperimentRealisationPlan $\sqsubseteq$ ExperimentalProtocol
ExperimentRealisationPlan $\sqsubseteq \exists realisesProposal$.SelectedExperiment
ExperimentRealisationPlan $\sqsubseteq \exists hasSession$.ProtocolSession

ProtocolSession One trip of the plate out of the incubator at a scheduled culture time.

$$\begin{aligned}
&\mathsf{ProtocolSession} \sqsubseteq \mathsf{dul{:}Description}\\
&\mathsf{ProtocolSession} \sqsubseteq \exists \mathit{scheduledAt_h}.\mathsf{xsd{:}double}\\
&\mathsf{ProtocolSession} \sqsubseteq \exists \mathit{hasStep}.\mathsf{LabActionDesignator}
\end{aligned}$$

LabActionDesignator A designator of the realisation plan (RS ScientificActionDesignator).

$$\begin{aligned}
&\mathsf{LabActionDesignator} \sqsubseteq \mathsf{ScientificActionDesignator}\\
&\mathsf{LabActionDesignator} \sqsubseteq \exists \mathit{stepIndex}.\mathsf{xsd{:}integer}
\end{aligned}$$

MaterialStepDesignator What happens to the material: dose well w, image well w.

$$\begin{aligned}
&\mathsf{MaterialStepDesignator} \sqsubseteq \mathsf{LabActionDesignator}\\
&\mathsf{MaterialStepDesignator} \sqsubseteq \mathsf{MaterialScientificDesignator}\\
&\mathsf{MaterialStepDesignator} \sqsubseteq \exists \mathit{refinesInto}.\mathsf{SceneStepDesignator}\\
&\mathsf{MaterialStepDesignator} \sqcap \mathsf{SceneStepDesignator} \sqsubseteq \bot
\end{aligned}$$

ChallengeDoseDesignator Dose challenge.

$$\begin{aligned}
&\mathsf{ChallengeDoseDesignator} \sqsubseteq \mathsf{MaterialStepDesignator}\\
&\mathsf{ChallengeDoseDesignator} \sqsubseteq \exists \mathit{doseAU}.\mathsf{xsd{:}double}
\end{aligned}$$

RescueDoseDesignator Dose rescue.

$$\begin{aligned}
&\mathsf{RescueDoseDesignator} \sqsubseteq \mathsf{MaterialStepDesignator}\\
&\mathsf{RescueDoseDesignator} \sqsubseteq \exists \mathit{doseAU}.\mathsf{xsd{:}double}
\end{aligned}$$

ImageWellDesignator Image a well.

$$\begin{aligned}
&\mathsf{ImageWellDesignator} \sqsubseteq \mathsf{MaterialStepDesignator}\\
&\mathsf{ImageWellDesignator} \sqsubseteq \exists \mathit{concernsWell}.\mathsf{Well}
\end{aligned}$$

SceneStepDesignator Bound to the scene: device, robot, poses.

$$\begin{aligned}
&\mathsf{SceneStepDesignator} \sqsubseteq \mathsf{LabActionDesignator}\\
&\mathsf{SceneStepDesignator} \sqsubseteq \mathsf{SceneGroundedDesignator}\\
&\mathsf{SceneStepDesignator} \sqsubseteq \exists \mathit{boundToDevice}.\mathsf{LabArtifact}\\
&\mathsf{SceneStepDesignator} \sqcap \mathsf{MaterialStepDesignator} \sqsubseteq \bot
\end{aligned}$$

PlateTransferDesignator Move the plate between two seat poses with one robot (one leg of a route).

$$\begin{aligned}
&\mathsf{PlateTransferDesignator} \sqsubseteq \mathsf{SceneStepDesignator}\\
&\mathsf{PlateTransferDesignator} \sqsubseteq \exists \mathit{performedByRobot}.\mathsf{CellRobot}\\
&\mathsf{PlateTransferDesignator} \sqsubseteq \exists \mathit{boundToPose}.\mathsf{PlateSeatPose}\\
&\mathsf{PlateTransferDesignator} \sqsubseteq \exists \mathit{refinesInto}.\mathsf{MotionStepDesignator}
\end{aligned}$$

DeviceOperationDesignator A device operation (dose, image, present, store).

$$\begin{aligned}
&\mathsf{DeviceOperationDesignator} \sqsubseteq \mathsf{SceneStepDesignator}\\
&\mathsf{DeviceOperationDesignator} \sqsubseteq \exists \mathit{boundToDevice}.\mathsf{LabAutomationDevice}\\
&\mathsf{DeviceOperationDesignator} \sqsubseteq \exists \mathit{refinesInto}.\mathsf{DeviceCommandDesignator}
\end{aligned}$$

MotionStepDesignator A robot body-motion primitive (RS BodyMotionDesignator) bound to a pose.

$$\begin{aligned}
&\mathsf{MotionStepDesignator} \sqsubseteq \mathsf{LabActionDesignator}\\
&\mathsf{MotionStepDesignator} \sqsubseteq \mathsf{BodyMotionDesignator}\\
&\mathsf{MotionStepDesignator} \sqsubseteq \exists \mathit{performedByRobot}.\mathsf{CellRobot}\\
&\mathsf{MotionStepDesignator} \sqsubseteq \exists \mathit{primitiveName}.\mathsf{xsd{:}string}\\
&\mathsf{MotionStepDesignator} \sqsubseteq \exists \mathit{estimatedDuration_s}.\mathsf{xsd{:}double}\\
&\mathsf{MotionStepDesignator} \sqcap \mathsf{MaterialStepDesignator} \sqsubseteq \bot
\end{aligned}$$

ApproachMotion Move to the approach pose above a seat.

$$\begin{aligned}
&\mathsf{ApproachMotion} \sqsubseteq \mathsf{MotionStepDesignator}\\
&\mathsf{ApproachMotion} \sqsubseteq \exists \mathit{boundToPose}.\mathsf{TwinPose}
\end{aligned}$$

InsertMotion Descend to the seat pose (compliant).

$$\begin{aligned}
&\mathsf{InsertMotion} \sqsubseteq \mathsf{MotionStepDesignator}\\
&\mathsf{InsertMotion} \sqsubseteq \exists \mathit{boundToPose}.\mathsf{TwinPose}
\end{aligned}$$

GraspMotion Close the gripper on the plate.

$$\mathsf{GraspMotion} \sqsubseteq \mathsf{MotionStepDesignator}$$
$$\mathsf{GraspMotion} \sqsubseteq \exists \mathit{boundToPose}.\mathsf{TwinPose}$$

LiftMotion Lift the plate off the seat.

$$\mathsf{LiftMotion} \sqsubseteq \mathsf{MotionStepDesignator}$$
$$\mathsf{LiftMotion} \sqsubseteq \exists \mathit{boundToPose}.\mathsf{TwinPose}$$

TransitMotion Planned joint-space / rail transit between approach poses.

$$\mathsf{TransitMotion} \sqsubseteq \mathsf{MotionStepDesignator}$$
$$\mathsf{TransitMotion} \sqsubseteq \exists \mathit{boundToPose}.\mathsf{TwinPose}$$

PlaceMotion Compliant placement into the target seat.

$$\mathsf{PlaceMotion} \sqsubseteq \mathsf{MotionStepDesignator}$$
$$\mathsf{PlaceMotion} \sqsubseteq \exists \mathit{boundToPose}.\mathsf{TwinPose}$$

ReleaseMotion Open the gripper.

$$\mathsf{ReleaseMotion} \sqsubseteq \mathsf{MotionStepDesignator}$$
$$\mathsf{ReleaseMotion} \sqsubseteq \exists \mathit{boundToPose}.\mathsf{TwinPose}$$

RetractMotion Retract to the approach pose (leaves device airspace).

$$\mathsf{RetractMotion} \sqsubseteq \mathsf{MotionStepDesignator}$$
$$\mathsf{RetractMotion} \sqsubseteq \exists \mathit{boundToPose}.\mathsf{TwinPose}$$

ToolCoupleMotion Couple an exchangeable tool at the magazine.

$$\mathsf{ToolCoupleMotion} \sqsubseteq \mathsf{MotionStepDesignator}$$
$$\mathsf{ToolCoupleMotion} \sqsubseteq \exists \mathit{boundToPose}.\mathsf{TwinPose}$$

DeviceCommandDesignator A device-level command (present, stow, tip pick, aspirate, dispense, eject, load, image, store, QC).

$$\mathsf{DeviceCommandDesignator} \sqsubseteq \mathsf{LabActionDesignator}$$
$$\mathsf{DeviceCommandDesignator} \sqsubseteq \mathsf{ScientificActionDesignator}$$
$$\mathsf{DeviceCommandDesignator} \sqsubseteq \exists \mathit{commandsDevice}.\mathsf{LabAutomationDevice}$$
$$\mathsf{DeviceCommandDesignator} \sqsubseteq \exists \mathit{primitiveName}.\mathsf{xsd{:}string}$$
$$\mathsf{DeviceCommandDesignator} \sqcap \mathsf{MaterialStepDesignator} \sqsubseteq \bot$$

SafetyStepDesignator A step inserted to discharge a safety rule (environment wait, QC calibration, interlock check).

$$\mathsf{SafetyStepDesignator} \sqsubseteq \mathsf{DeviceCommandDesignator}$$
$$\mathsf{SafetyStepDesignator} \sqsubseteq \exists \mathit{dischargesRule}.\mathsf{LabSafetyRule}$$

PlanDerivation One run of the derivation (answer-set program over the ontology's ABox and the EIG candidates).

$$\mathsf{PlanDerivation} \sqsubseteq \mathsf{dul{:}Description}$$
$$\mathsf{PlanDerivation} \sqsubseteq \exists \mathit{derivesPlan}.\mathsf{ExperimentRealisationPlan}$$
$$\mathsf{PlanDerivation} \sqsubseteq \exists \mathit{appliesAxiomSet}.\mathsf{DerivationAxiomSet}$$

DerivationAxiomSet The L3 axioms E2M1-E2M8 (L3-logical-axiomatization/axioms/experiment-to-motion-axioms.md).

$$\mathsf{DerivationAxiomSet} \sqsubseteq \mathsf{dul{:}InformationObject}$$
$$\mathsf{DerivationAxiomSet} \sqsubseteq \exists \mathit{sourceDocument}.\mathsf{xsd{:}string}$$

RealisedInformationGain KL(p(M|D,y) ‖ p(M|D)) after executing the plan.

$$\mathsf{RealisedInformationGain} \sqsubseteq \mathsf{dul{:}InformationObject}$$
$$\mathsf{RealisedInformationGain} \sqsubseteq \exists \mathit{realisedGain_bits}.\mathsf{xsd{:}double}$$
$$\mathsf{RealisedInformationGain} \sqsubseteq \exists \mathit{about}.\mathsf{ExperimentRealisationPlan}$$